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SPOT MARKET
HANDBOOK
HOW AGRICULTURAL PRICES
ACTUALLY WORK
A PRACTICAL GUIDE TO PHYSICAL MARKETS,
PRICE DISCOVERY AND COMMODITY INDICES
UNDERSTAND REAL MARKETS
FOLLOW PRICE DISCOVERY
MANAGE BASIS AND RISK
BUILD MARKET INTELLIGENCE
MAKE BETTER DECISIONS
GLOBAL EXPERIENCE.
REAL MARKETS.
BETTER DECISIONS.
DATA • BENCHMARKS • INDICES • ANALYTICS • INFRASTRUCTURE

SPOT MARKET HANDBOOK

How Agricultural Prices Actually Work

A Practical Guide to Physical Markets, Price Discovery,
Commodity Indices and Market Intelligence

Prepared within the development of:

1D3X, MN7R, Cr0pto

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Spot Market Handbook
How Agricultural Prices Actually Work

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Project websites:

1d3x.com
mn7r.com
cr0pto.com

ABOUT THIS BOOK

This handbook was created to explain how physical commodity markets function beyond futures exchanges, headlines and daily price reports.

While many market participants follow benchmark prices, relatively few understand how those prices emerge from logistics, basis relationships, liquidity conditions, respondent networks and physical market activity.

The purpose of this book is to make these mechanisms easier to understand.

Although examples from Ukraine and the Black Sea region are used throughout the book, the concepts discussed here apply broadly across agricultural commodity markets worldwide.

The book is intended for traders, brokers, exporters, processors, analysts, investors, students and anyone interested in how physical commodity markets actually work.

DISCLAIMER

This publication is provided solely for educational and informational purposes.

Nothing in this book should be interpreted as investment advice, trading advice, legal advice, financial advice or a recommendation to buy or sell any commodity, derivative, security or financial instrument.

Examples, illustrations, scenarios and market situations presented in this book are intended to explain market concepts and methodologies. They do not constitute forecasts of future market behavior.

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All charts, diagrams, figures and illustrations included in this publication are provided for explanatory purposes.

Unless explicitly stated otherwise, they are illustrative representations of market concepts and methodologies rather than historical market records.

Illustrations may simplify complex market relationships in order to improve understanding and should not be interpreted as precise representations of actual market events.

Examples involving prices, basis relationships, logistics flows, benchmark construction, volatility, spreads or market infrastructure are intended to demonstrate concepts rather than document specific transactions.

ABOUT THE PROJECTS

This book is part of a broader effort to improve transparency, benchmark infrastructure and market intelligence within agricultural commodity markets.

- **1D3X** develops index infrastructure and benchmark frameworks for local commodity markets.
- **MN7R** focuses on market intelligence, commodity analytics and data-driven decision support.
- **Cr0pto** explores future applications of commodity indices in programmable and tokenized market environments.

Together, these initiatives aim to transform fragmented market information into structured, transparent and actionable market intelligence.

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FOREWORD

Why Physical Markets Need a New Language

Most commodity markets look simpler from the outside than they actually are.

A price appears on a screen. A futures contract moves up or down. A report mentions a benchmark. An analyst says the market is bullish, bearish, tight or oversupplied. But behind every quoted number there is a physical world.

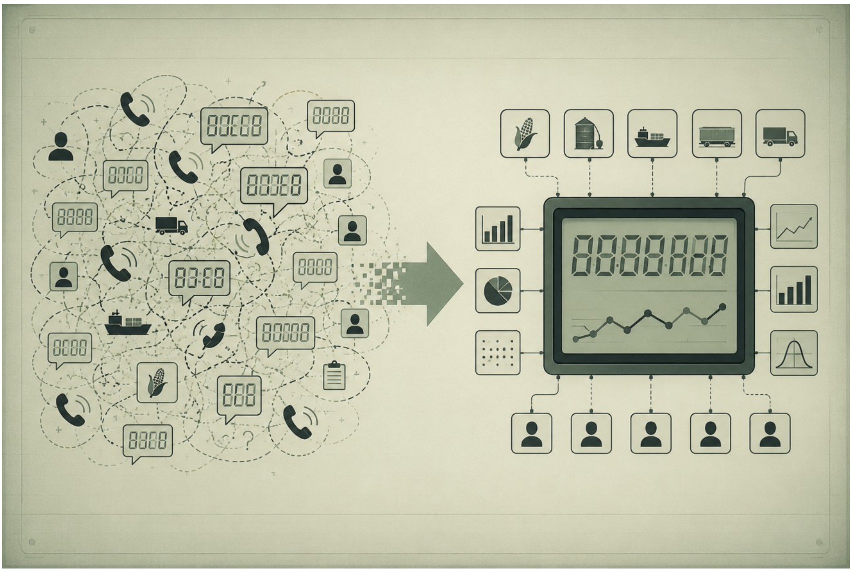
Grain has to be grown, stored, transported, inspected, financed, shipped, processed or exported. A price is never only a number. It is a compressed expression of logistics, liquidity, timing, quality, risk, geography, credit, execution pressure and market confidence. This is why physical agricultural markets often behave differently from global futures markets.

A futures price may rise while a local export price weakens. A regional basis may collapse even when global demand looks strong. A port restriction, a freight shock, a currency move or a sudden change in local liquidity can reshape the economics of the physical market within days.

For traders, brokers, exporters, processors and analysts, this is not theory. It is daily work.

The problem is that much of this daily work still happens in fragmented information environments: phone calls, private chats, spreadsheets, broker messages, internal reports and partial market views. Some participants see a deep flow of bids, offers and executed trades. Others see only a few quotes and market rumors.

This creates a structural information gap.



This diagram illustrates the central challenge of physical commodity markets. Valuable information exists throughout the market, but it is scattered across participants, regions, logistics chains and commercial relationships.

A well-designed spot index does not attempt to simplify that reality. Its role is more practical. It organizes fragmented observations into a common reference that market participants can understand, compare and discuss.

The objective is not to remove complexity. The objective is to make complexity readable.

A good index helps convert fragmented market signals into a shared reference point. It gives participants a way to discuss price movement, basis changes, liquidity conditions and market structure using a common language.

This book was written to explain that language. It is not a textbook on derivatives. It is not a manual for speculative trading. It is not a theoretical essay on commodity economics. It is a practical guide to how agricultural prices actually work in physical markets.

The handbook explains why local prices differ from futures prices, how basis risk emerges, why FOB and CPT prices tell different stories, how respondent-based indices are built, why median-based methodology matters, why thin markets require caution, and how daily spot indices can evolve into market intelligence infrastructure.

The Ukrainian market and the Black Sea region appear in this book as important examples, because they show many of these mechanisms clearly: logistical stress, export route changes, basis volatility, fragmented OTC price discovery and the growing need for transparent local benchmarks. But the logic is not limited to Ukraine.

The same questions appear in many agricultural markets:

- *What is the real local price?*
- *Which basis matters?*
- *How liquid is the market today?*

- *Which quote reflects executable value?*
- *Is the market moving, or are we seeing noise?*
- *Can a benchmark be trusted?*
- *How can market data become useful for analytics, contracts, APIs and future trading infrastructure?*

These questions are central to the work behind 1D3X, MN7R and Cr0pto.

- **1D3X** approaches commodity markets as networks of local benchmarks that can become part of global market infrastructure.
- **MN7R** focuses on market intelligence and decision support for physical commodity workflows.
- **Cr0pto** explores how commodity indices may eventually support programmable and tokenized market models.

Together, these projects start from a simple premise: Before a market can become more digital, more automated or more investable, it must first become more readable.

That is where spot indices begin.

Not as abstract numbers.
Not as decorative market statistics.

But as a practical language that allows fragmented markets to communicate with themselves.

- A way to see structure where there was noise.
- A way to compare local realities.
- A way to turn daily market signals into intelligence.

This handbook is an introduction to that logic.

Chapter 1: The New Language of Commodity Markets

Every market develops its own language.

Not a language of words, but a language of signals, references, assumptions and shared meanings.

Participants who understand that language can navigate complexity efficiently. Those who do not often find themselves overwhelmed by information while still lacking understanding.

Stock markets speak in earnings, multiples and volatility.

Bond markets speak in yields and duration.

Foreign exchange markets speak in rates, spreads and flows.

Physical commodity markets have their own vocabulary as well.

- *Basis*.
- *FOB*.
- *CPT*.
- *Liquidity*.
- *Execution*.
- *Spread*.
- *Freight*.
- *Market depth*.

- *Respondents.*
- *Benchmark.*

For experienced traders, these terms become second nature. They are part of everyday conversations, negotiations and market decisions.

For newcomers, however, commodity markets often appear confusing.

- A buyer asks for a FOB indication.
- A seller responds with a CPT number.
- A broker references basis movement.
- An analyst discusses a benchmark.
- A futures trader points to Chicago.

Meanwhile, a processor is watching local cash prices. Everyone appears to be discussing the same commodity.

Yet everyone seems to be speaking a different language.

The reality is that they are.

One of the biggest misconceptions in commodity trading is the idea that a market has a single price.

In practice, almost every commodity has multiple prices at the same moment.

- A farmer may see one value.
- An exporter may see another.
- A processor may calculate a third.

- A futures exchange may imply a fourth.

Each of these prices can be correct within its own context.

The challenge is rarely the absence of information.

The challenge is abundance without structure.

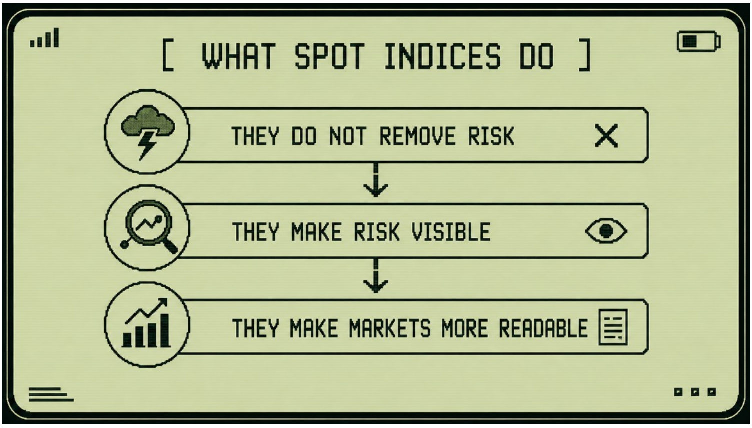
Modern commodity markets generate more information than any individual participant can realistically process. The problem is no longer access. The problem is interpretation.

- Quotes arrive from multiple sources.
- Different companies report different levels.
- Market participants interpret the same events differently.
- Local logistics create regional distortions.
- Liquidity shifts throughout the day.
- Rumors compete with verified information.
- Without structure, information becomes noise.

The more fragmented the market becomes, the harder it is to distinguish a meaningful signal from random variation. This is where benchmarks become important.

A benchmark does not reduce market complexity. It provides a framework for understanding it.

A spot index does not remove risk from the market. Its value is more practical: it makes risk visible, comparable and easier to discuss. One Commodity — Multiple Prices.



Instead of forcing market participants to compare dozens of isolated quotes, a benchmark creates a common reference point.

A benchmark is not the market itself. It is a representation of the market center at a specific moment in time.

This distinction is important.

No benchmark can perfectly capture every transaction.

No benchmark can reflect every participant's individual reality.

Its purpose is different. Its purpose is to create comparability. Once a benchmark exists, participants gain a shared frame of reference.

Traders can discuss whether prices are above or below the benchmark.

- Analysts can measure trends.
- Exporters can compare routes.

- Processors can evaluate local competitiveness.
- Investors can observe market behavior.

The benchmark becomes part of the market conversation.

In mature commodity markets, benchmarks eventually become embedded in contracts, analytics systems, reporting frameworks and risk management processes.

Over time, they stop being viewed as publications and start functioning as infrastructure. This transformation rarely happens overnight.

It begins with something much simpler.

A market decides to create a shared language. A language capable of translating thousands of fragmented signals into something participants can understand collectively.

That is the role of a spot index. It turns information into structure. It turns structure into comparability. And it creates the foundation for market intelligence.

The chapters that follow explore how this language is constructed. We begin with the relationship between local physical markets and global futures markets. From there, we move through delivery economics, benchmark construction, respondent methodology, market intelligence and the infrastructure layers that increasingly support modern commodity trading.

Understanding starts with a simple question: *If global futures prices are visible to everyone, why do local markets still need their own benchmarks?*

Chapter 2: Why Local Markets Are Not Futures Markets

One of the most persistent misconceptions in commodity trading is the belief that futures markets and physical markets describe the same reality.

They are connected. But they are not identical.

A futures contract and a physical cargo may refer to the same commodity, yet they often respond to different forces.

This distinction becomes especially important when markets experience volatility, logistical disruption or regional imbalances.

To understand why spot indices exist, we first need to understand the difference between futures markets and physical markets.

A futures market is designed to facilitate price discovery, risk transfer and hedging. It concentrates liquidity.

Participants can enter and exit positions rapidly.

Prices respond continuously to expectations about future supply and demand.

- Weather forecasts.
- Government reports.
- Interest rates.
- Currency movements.
- Macroeconomic sentiment.

- Speculative flows.
- Institutional positioning.

All of these can influence futures prices.

A futures contract therefore reflects the market's collective expectations about the future. A physical market operates differently. Physical participants must move real commodities through real supply chains.

They deal with storage constraints, freight costs, transportation bottlenecks, quality differences, local demand, infrastructure limitations and execution risk. Their concerns are often immediate rather than theoretical.

- *Can the cargo be loaded?*
- *Is transportation available?*
- *What is the local demand today?*
- *How much inventory exists in the region?*
- *Can the buyer actually execute the transaction?*

These questions do not disappear simply because futures prices change. As a result, futures markets and physical markets can move in the same direction while behaving very differently. They can even move in opposite directions.

Imagine a situation where weather concerns in a major producing region trigger a rally in futures markets.

- Investors buy contracts.
- Funds increase exposure.

- Analysts revise supply forecasts.
- Futures prices rise.

At the same time, a local export market may face logistical congestion, limited storage capacity or weak buying activity.

Physical buyers become cautious. Export demand slows. Basis weakens. Local cash prices stagnate or decline.

The futures market is reacting to expectations.

The physical market is reacting to execution realities.

Both are measuring value. They are simply measuring different dimensions of it.

This distinction explains why professional commodity traders rarely monitor futures prices alone.

A futures price tells part of the story.

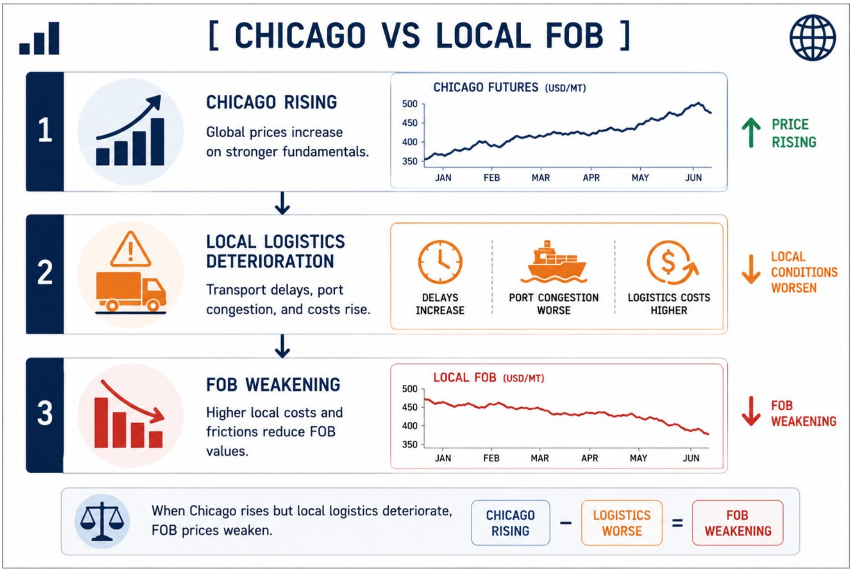
The physical market provides the rest.

The bridge between these two worlds is basis.

Basis represents the difference between a futures reference and a local physical market value. It reflects transportation costs, regional supply conditions, local demand, logistics, liquidity and execution risk.

FOB and CPT are not just contract terms. They describe different economic positions in the delivery chain, with different costs, responsibilities and risks.

FOB Price Calculation / Chicago Futures to FOB Price.



Basis can be understood as the physical market's response to the futures market. It reveals where local reality diverges from global expectation.

This is one of the reasons local spot indices are valuable.

A futures exchange may provide a global benchmark. But it cannot measure every regional market.

It cannot capture every export corridor. It cannot monitor every local logistics constraint. It cannot describe every physical basis.

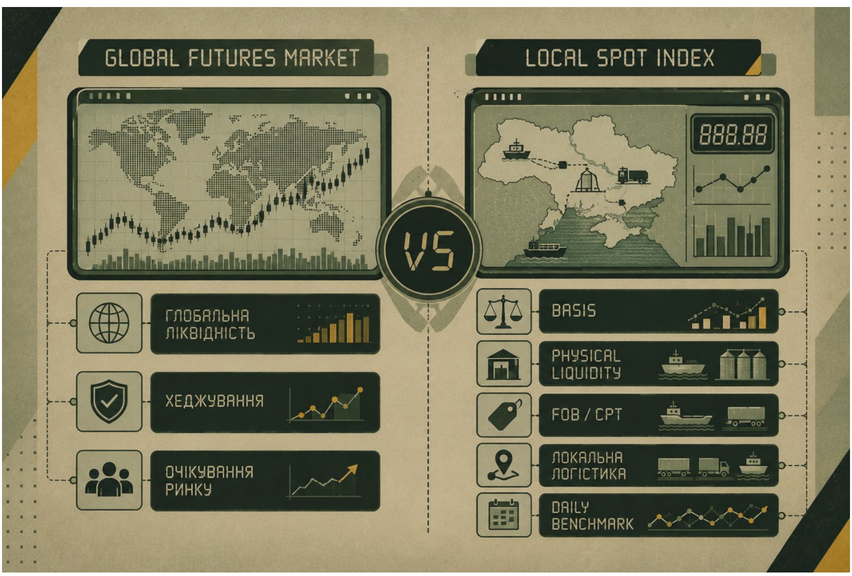
Local spot indices fill this gap.

- They provide a structured view of physical market conditions.

- They transform scattered local information into an organized benchmark.
- They help traders, exporters, processors and analysts understand what is happening beyond the futures screen.

The FOB price can be understood as a local physical expression of a global reference price adjusted by basis, logistics, port liquidity and freight.

Indicative vs Bid vs Offer vs Executed Trade.



This becomes increasingly important as commodity markets grow more complex.

Global benchmarks remain essential. But global benchmarks alone are no longer enough.

Modern commodity markets require both perspectives: *the global view provided by futures markets; and the local view provided by physical market benchmarks*.

Understanding how these two layers interact is the foundation of modern commodity market intelligence. Only after this distinction becomes clear can we begin to understand how physical prices are actually formed.

The next question naturally follows:

If futures markets cannot fully describe local reality, how exactly are physical market prices formed?

Chapter 3: FOB, CPT and the Physics of Delivery

Commodity markets often appear simpler than they are.

A price exists.

A buyer exists.

A seller exists.

From a distance, the transaction seems straightforward.

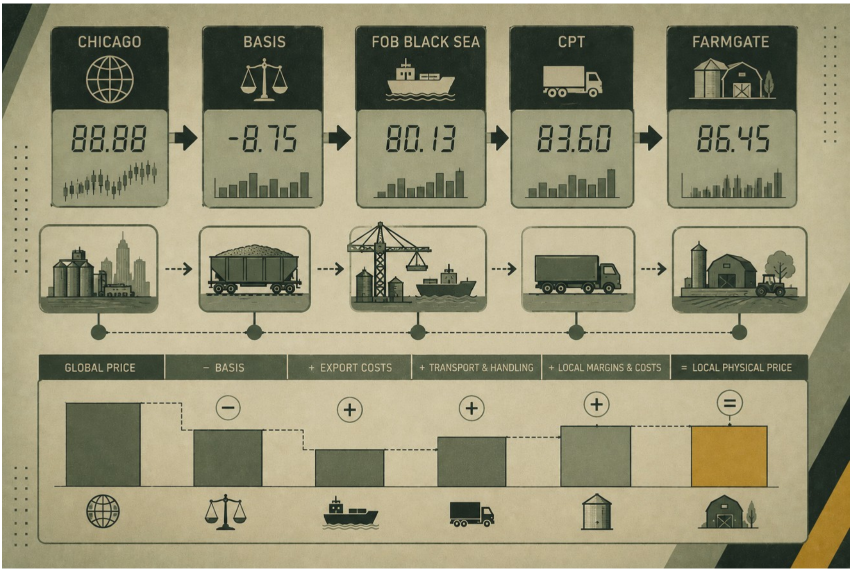
In reality, every physical commodity transaction sits inside a logistics system, a risk system and a delivery system. The commodity itself may remain unchanged, but the economics surrounding it can change dramatically from one location, route or delivery basis to another.

Commodities do not exist in abstract markets. They exist in physical supply chains.

A cargo of grain, coffee, sugar, soybeans or vegetable oil is not simply a financial asset. It must be moved, stored, handled, loaded, inspected, delivered and financed.

Every one of these activities has a cost. Every one introduces risk. Every one affects the final price.

This is why physical markets rarely have a single universally accepted value. Instead, they produce several related prices, each connected to a specific delivery point, route, responsibility and execution condition.



Two of the most common concepts used in agricultural trade are FOB and CPT. Understanding them is essential for understanding how local commodity prices are formed.

FOB stands for Free On Board. Under FOB terms, the seller is responsible for delivering the commodity onto a vessel at a designated export port. Once the cargo is loaded, responsibility transfers to the buyer. The FOB price therefore reflects everything required to move the commodity from its origin to the export vessel.

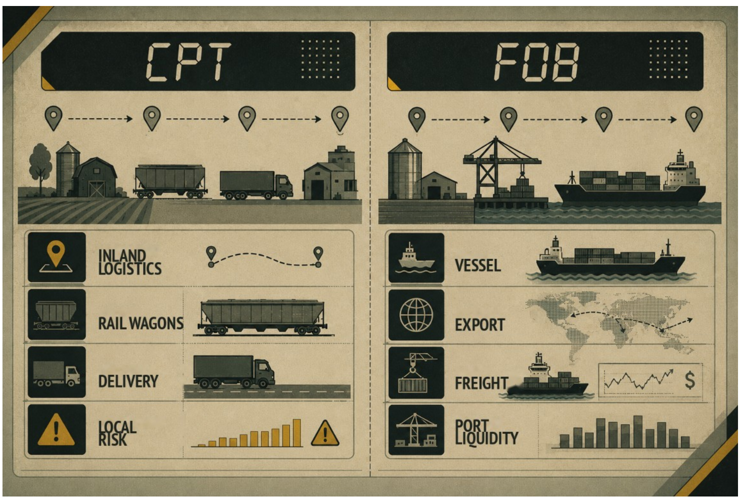
CPT stands for Carriage Paid To. Under CPT terms, the seller is responsible for transporting the commodity to a specified destination. In agricultural markets, this may mean CPT port, CPT processor, CPT elevator or CPT terminal. A CPT price includes transportation costs up to that location.

Neither FOB nor CPT is inherently more important. They simply answer different commercial questions.

FOB asks: *what is the value of the commodity at the export vessel?*

CPT asks: *what is the value of the commodity at a specified destination?*

Understanding which question is being answered is often more important than the number itself.



This is where many market participants encounter their first real lesson in basis economics.

A commodity does not become more valuable simply because it moves. Its price changes because the economics of delivery change.

A truck moving grain to an inland processor faces different costs than a cargo destined for export. A vessel loading at a deep-water port faces different constraints than a truck delivering to a local crushing facility.

As a result, a commodity rarely has a single universally correct price. Instead, it exists as a family of related prices connected by logistics, distance, infrastructure, risk and delivery obligations.

A farmer may see a farmgate price.

A trader may quote CPT.

An exporter may negotiate FOB.

A processor may work from an entirely different basis.

None of these prices are necessarily wrong. They simply reflect different positions within the supply chain. This is one of the first lessons every commodity professional eventually learns.

The question is rarely: *"What is the price?"*

The more useful question is: *"Which price?"*

Or, more precisely: *"What is the price where?"*

The delivery point matters.

The route matters.

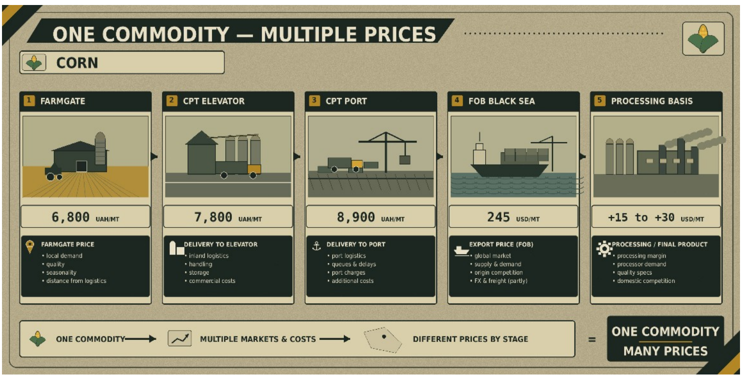
The infrastructure matters.

The execution conditions matter.

The market is pricing not only the commodity itself, but also the process of moving it.

This becomes even more visible when logistics are disrupted. A transportation bottleneck, a port closure, a shortage of rail capacity or a sudden increase in freight costs can change physical prices dramatically even when global futures prices remain relatively stable.

The commodity has not changed. The economics of delivery have.



This is why experienced commodity professionals rarely ask: *“What is the price?”*

Instead, they ask: *“What is the price where?”*

- The delivery point matters.
- The route matters.
- The infrastructure matters.
- The execution conditions matter.
- The market is pricing not only the commodity itself, but also the process of moving it.

This becomes even more visible when logistics are disrupted.

- A transportation bottleneck.
- A port closure.
- A shortage of rail capacity.
- A sudden increase in freight costs.

Any of these can change physical prices dramatically even when global futures prices remain relatively stable. The commodity has not changed. The economics of delivery have.

This is one reason why physical commodity markets often appear confusing to outsiders. Two market participants may discuss the same product while quoting completely different prices. Yet both may be correct. They are simply observing different parts of the delivery chain.

Case 01. Port Restriction and Basis Widening

In physical agricultural markets, price does not always move because the commodity itself has become more scarce or more abundant.

Sometimes the strongest price movement comes from a change in logistics.

Imagine a Black Sea export market where part of the port capacity becomes temporarily unavailable. The reason *may be a security risk, vessel congestion, technical restrictions, overloaded terminals, documentation delays or limited access to loading infrastructure.*

On the global market, this event may not immediately change expectations for corn or wheat. Chicago futures may remain stable. International demand may not change. The global balance sheet may look almost the same as it did the day before. But for the local physical market, the economics change immediately.

Exporters become more cautious with procurement prices. Freight and handling costs increase. Some volumes are redirected toward alternative routes. Sellers who previously focused on one delivery basis now have to compare several options: Greater Odesa, Danube ports, western border crossings, Baltic routes or inland processing demand. At this point, the basis begins to widen.

This means that the difference between the global reference price and the local physical price becomes larger.

For a trader, this is critical. If the trader looks only at the outright price, the market may appear confusing. The commodity has not changed. Futures may not have collapsed. Demand may still exist. Yet the local export value becomes weaker. But if the trader follows the local spot index, FOB/CPT spreads and basis movement, the picture becomes clearer.

At moments like this, the market is not primarily evaluating the commodity itself. It is evaluating the *efficiency, reliability and cost of moving that commodity through the supply chain.*

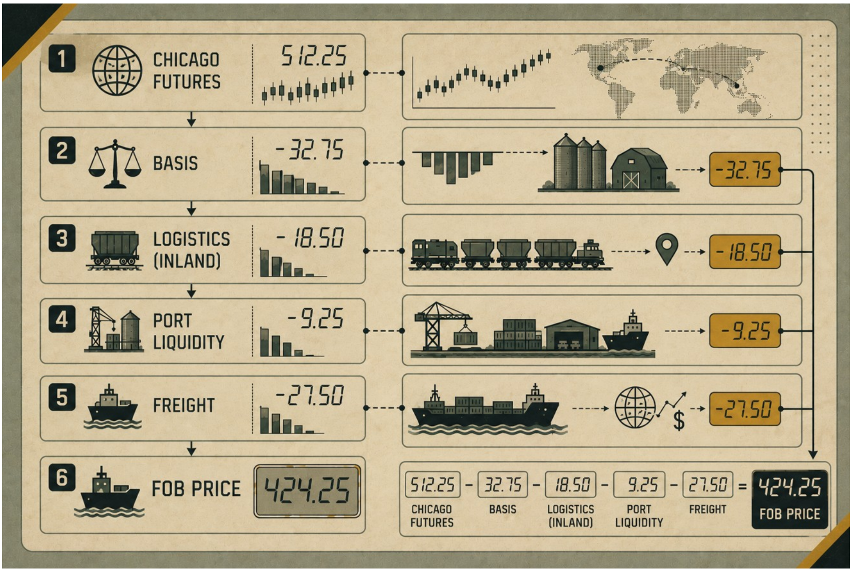
A port restriction does not always mean a market collapse. But it almost always changes route economics. It changes where liquidity concentrates, how exporters price risk, how sellers compare alternatives and how quickly cargo can be executed. This is exactly the kind of change a local spot index helps reveal. It does not simply tell the market that the price moved. It helps explain why the price moved.

This case brings us back to the practical value of delivery spreads. When logistics changes, the spread between delivery bases becomes one of the clearest signals in the physical market.

The spread between these prices contains valuable information.

- It tells us where costs are rising.
- Where liquidity is concentrating.
- Where demand is strengthening.
- Where infrastructure is under pressure.

In many cases, the spread itself becomes more informative than the outright price.



Modern commodity analytics therefore focuses not only on simple price levels, but also on basis relationships, delivery economics and route comparisons.

A spot index does more than publish a number. It provides a structured way to observe the economics of physical delivery.

Once we begin observing delivery economics, another question emerges: *How do traders determine whether a quoted price actually reflects the market?*

To answer it, we need to move from delivery economics to market behavior.

Prices do not appear automatically. They emerge through conversations, negotiations, bids, offers and transactions. That process is what we call the spot market.

Chapter 4: What a Spot Market Looks Like in Practice

Many people imagine a market as a continuous stream of prices. Numbers move every second. Charts update in real time. Participants observe a constantly changing screen. This description fits many futures markets. It does not fully describe most physical commodity markets.

A physical market operates differently. Instead of continuous trading activity, it often functions through periods of negotiation, execution and temporary liquidity. For this reason, a spot market is not simply a place where prices exist.

It is a place where transactions become possible. The distinction matters.

A quoted price and an executable price are not always the same thing.

- A market participant may indicate a level.
- A buyer may show interest.
- A seller may express willingness.

But until a transaction can actually be completed, the market has not fully confirmed that value.

This is why experienced commodity professionals pay close attention not only to prices, but also to market quality.

- *How many participants are active?*
- *How many bids exist?*

- *How many offers exist?*
- *How much liquidity is available?*
- *How realistic is execution?*

These questions are often more important than the price itself. To understand this process, we need to distinguish between four common types of market information.

Indicative

An indicative value is a market indication. It represents where a participant currently sees the market.

Indicatives help establish context. They help market participants begin discussions. However, an indicative is not necessarily executable. It is a signal. Not a transaction.

Bid

A bid is a buying interest. It represents the level at which a participant is prepared to purchase a commodity.

A bid is stronger than an indicative because it reflects a willingness to transact.

Yet even a bid may contain conditions: Volume. Timing. Quality. Location. Logistics.

The existence of a bid does not automatically guarantee execution.

Offer

An offer represents selling interest.

A seller indicates a willingness to sell at a particular level.

As with bids, offers help define the current market range. But an offer is not necessarily an executed trade.

It may simply reflect expectations or negotiating strategy.

Executed Trade

An executed trade represents the strongest form of market information.

A transaction has occurred. Buyer and seller have agreed. Execution has been confirmed. Yet even an executed trade requires interpretation.

One transaction does not always represent the broader market: An unusual volume. A distressed seller. A unique logistics situation. A special quality specification.

Any of these factors may make a transaction less representative of overall market conditions. This is one reason why professional benchmarks rely on samples rather than individual trades.

The goal is to identify the market center rather than a single event. A typical trading day illustrates why this distinction matters.

Early in the morning, traders review overnight developments.

- Futures markets.
- Currencies.
- Weather forecasts.
- Political developments.
- Logistics updates.
- Shipping conditions.

As the day progresses, bids and offers begin to emerge.

- Conversations take place.
- Information spreads.
- Participants test market interest.
- Many of these interactions never become trades.
- Nevertheless, they contribute to market discovery.

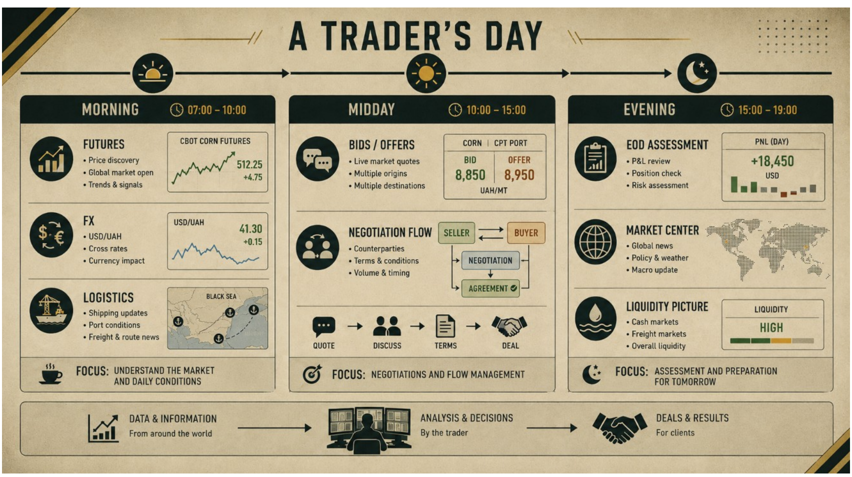
By late afternoon, the market begins to reveal a clearer structure.

- Liquidity concentrations become visible.
- Execution opportunities become clearer.
- Market participants gain a better understanding of the day's conditions.

This is why end-of-day market assessments are so important.

They capture not only isolated transactions, but the broader context surrounding them.

In physical commodity markets, the end of the day often provides a more reliable picture than any individual quote observed earlier.



Case 02. Danube Route vs Greater Odesa

On paper, grain export may look simple.

There is a commodity.
There is a buyer.
There is a port.
There is a price.

In the physical market, it is more complicated.

The same corn cargo may have different economics depending on the export route. A cargo moving through Greater Odesa and a cargo moving through the Danube corridor may belong to the same national export market, but they do not always have the same price, the same execution speed or the same risk profile.

Greater Odesa usually offers stronger scalability. It has deeper port infrastructure, larger vessel lots, more established export flows and a higher concentration of liquidity. When the route operates normally, it often forms a stronger and more representative export basis.

The Danube route follows a different logic. It can become critically important when access to larger ports is restricted or uncertain. But its economics often include more intermediate steps: inland logistics, transshipment, queues, smaller cargo lots, additional coordination and higher sensitivity to local delays.

The question is therefore not which route is "better" in an abstract sense.

The real question is: Which route provides an executable price today?

On a calm day, Greater Odesa may be cheaper, faster and more liquid. On a day of restrictions, risks or infrastructure pressure, the Danube route may become a necessary alternative, but with a different basis.

This is why a physical market cannot be reduced to one national price. Each route creates its own combination of cost, risk, time and liquidity.

For a trader, the difference is practical. A bid that looks attractive on one route may become less attractive after freight, waiting time, handling constraints or execution risk are included. An offer that appears expensive may be realistic if it reflects a route with better certainty.

A good spot index does not erase these differences. It organizes them.

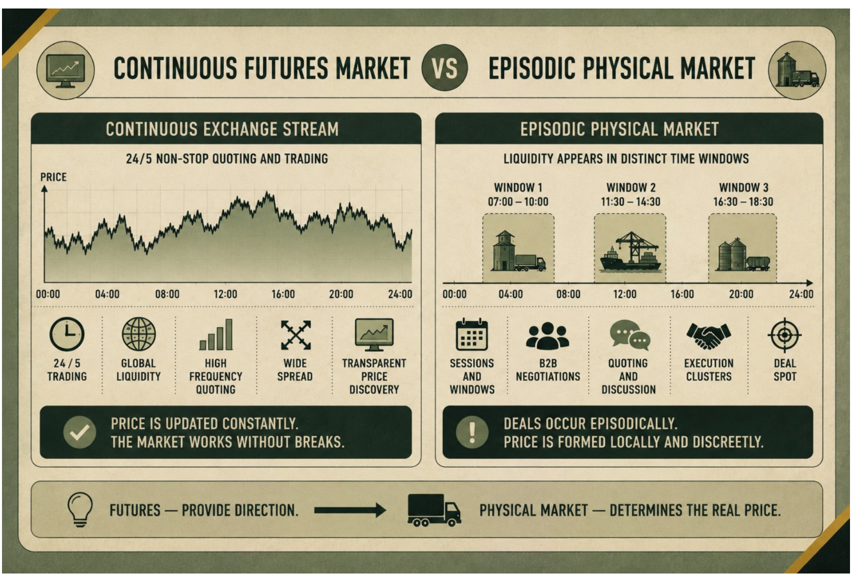
When a benchmark separately reflects delivery bases, routes and historical spreads, it helps the market see not just "the grain price", but the real economics of delivery.

The same route comparison also explains why liquidity in physical markets is episodic rather than continuous. Activity concentrates where execution is possible, not simply where a price is quoted.

Unlike futures markets, physical markets do not always provide continuous liquidity.

- Liquidity appears in clusters.
- Periods of intense activity may be followed by long periods of relative quiet.
- Certain hours become more important than others.
- Certain locations become more active than others.
- Certain participants temporarily dominate market flow.

These liquidity windows are a defining characteristic of physical commodity trading. Understanding them is essential for understanding spot indices.

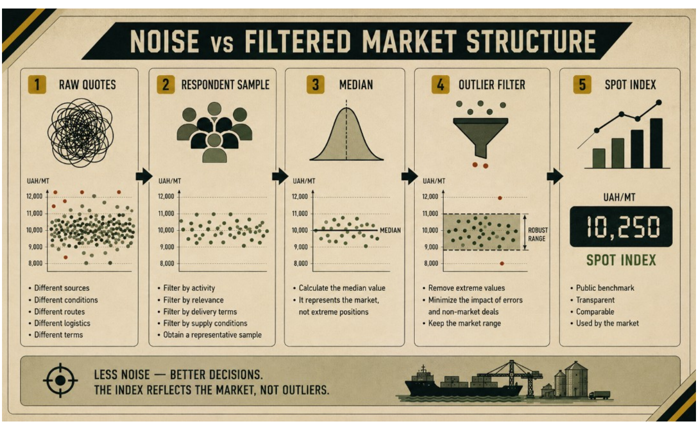


A spot index does not attempt to replicate every quote. It does not attempt to capture every conversation. Instead, it attempts to identify the center of market activity within a fragmented and episodic environment.

This is why spot indices have become increasingly important. The more fragmented a market becomes, the more valuable a structured reference point becomes.

A well-designed benchmark transforms scattered observations into something comparable.

- It provides continuity where markets provide fragments.
- It creates memory where markets create events.
- It turns individual signals into collective intelligence.



And that naturally raises another question: *If markets generate thousands of fragmented observations every day, how do we separate meaningful information from noise?*

Chapter 5: Why Markets Without Benchmarks Live in Information Noise

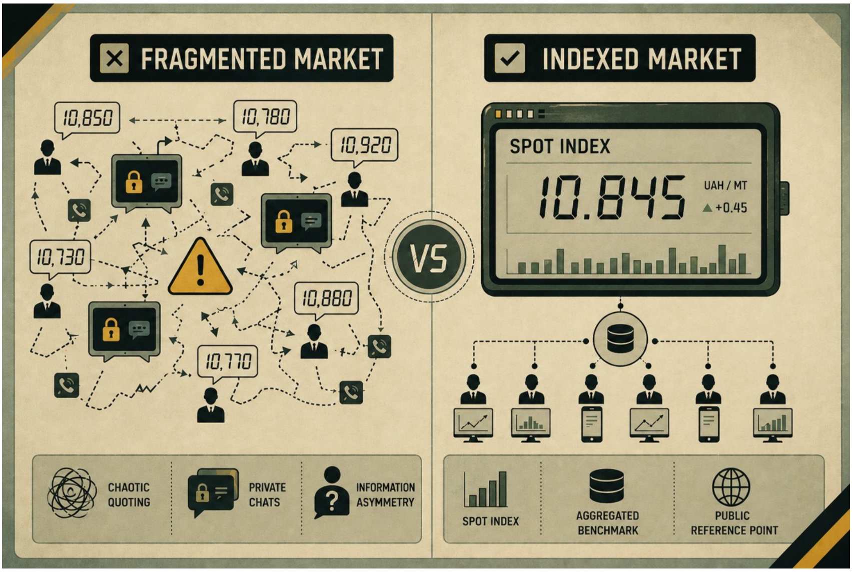
Most physical commodity markets generate enormous amounts of information every day.

- Phone calls.
- Broker messages.
- Private chats.
- Indicative quotes.
- Bids.
- Offers.
- Executed transactions.
- Shipping updates.
- Weather reports.
- Regional supply changes.
- Demand shifts.
- Logistics constraints.
- Currency fluctuations.

At first glance, this appears to be an advantage. More information should make markets easier to understand.

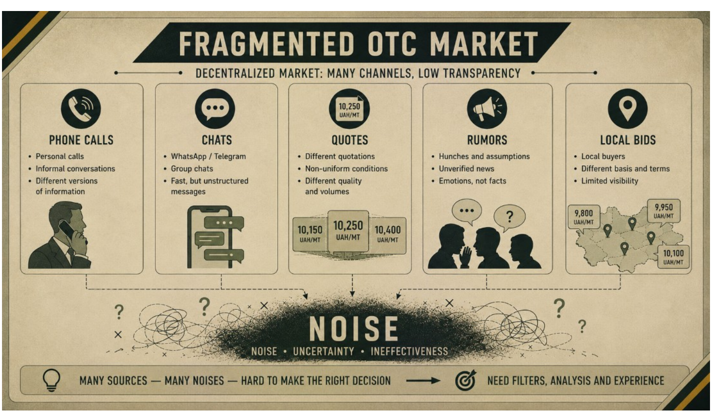
In practice, the opposite often happens.

The more fragmented a market becomes, the harder it becomes to distinguish information from noise.



This is especially true in OTC markets.

Unlike centralized exchanges, OTC markets do not provide a single consolidated view of activity.



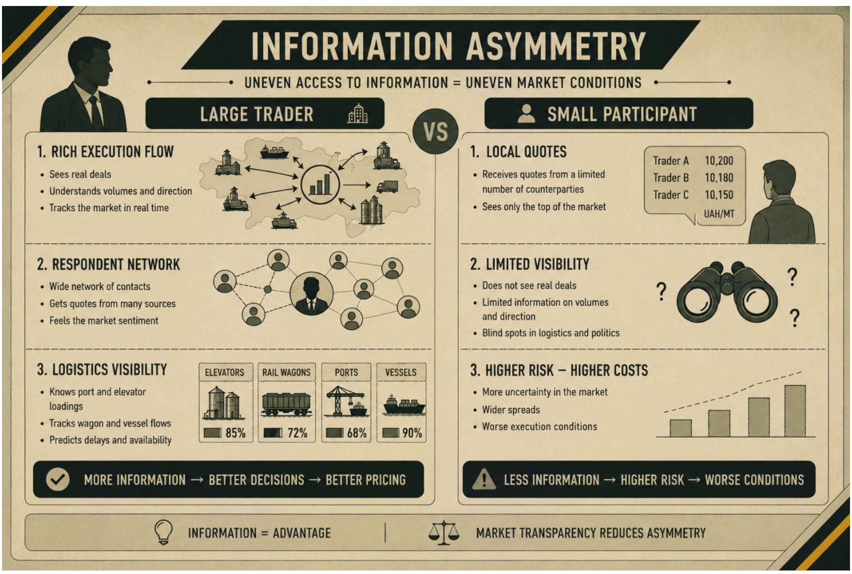
Information is distributed across many participants. Each participant sees only part of the picture.

A trader may observe one group of buyers.

An exporter may see another.

A processor may interact with a completely different set of counterparties.

No individual participant sees everything. This creates what economists call information asymmetry.



Some market participants simply have access to more information than others.

A large trading house may observe hundreds of daily interactions.

- A local buyer may see only a handful.

- A multinational exporter may monitor multiple regions simultaneously.
- A small participant may only know what happens within a narrow geographic area.

Neither participant is necessarily wrong. They are simply operating with different visibility. This creates a challenge. If every participant sees only fragments of reality, how does the market develop a common understanding of where prices actually are?

Historically, many markets relied on reputation and relationships. Experienced participants built informal networks. Information circulated through trusted contacts.

Over time, individuals developed an intuitive sense of market conditions.

This approach still works. To a point. The problem is scalability. As markets become larger, more global and more data-intensive, informal information networks become increasingly difficult to manage.

Noise begins to dominate.

- A single unusual quote may attract attention.
- An isolated transaction may be interpreted as a trend.
- A local logistics problem may be mistaken for a broader market movement.

Rumors can spread faster than verified information. The market becomes harder to read.

This is where benchmarks become valuable.

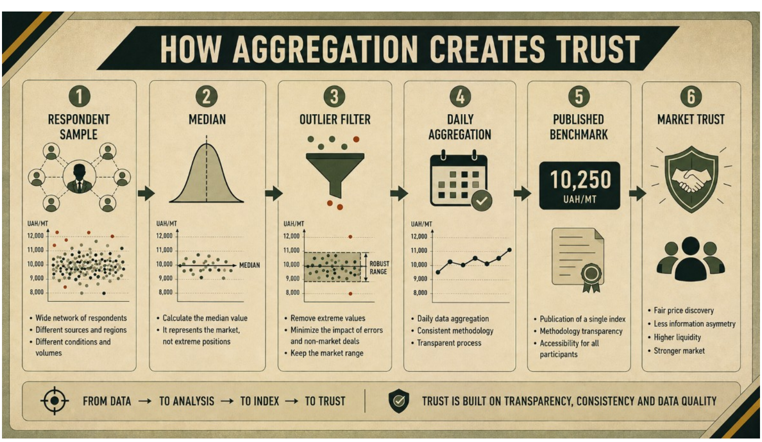
A benchmark does not eliminate information asymmetry. No benchmark can give every participant identical information. Its role is different.

A benchmark creates a shared reference layer. Instead of forcing participants to compare dozens of disconnected observations, it provides a common point of comparison.

This changes how markets communicate. Without a benchmark, participants often argue about individual prices.

With a benchmark, participants discuss deviations from a common reference.

- The conversation becomes more structured.
- The market becomes more comparable.
- The benchmark acts as a translation layer between fragmented observations and collective understanding.



This function becomes even more important in OTC markets. OTC markets often have deep liquidity.

But they also have fragmented visibility.

- Execution remains private.
- Negotiations remain private.
- Transactions may remain confidential.
- A benchmark does not require this information to become public.

Instead, it aggregates information while preserving confidentiality. This distinction is crucial.

Transparency does not require disclosure of every transaction. Transparency requires the ability to understand the market.

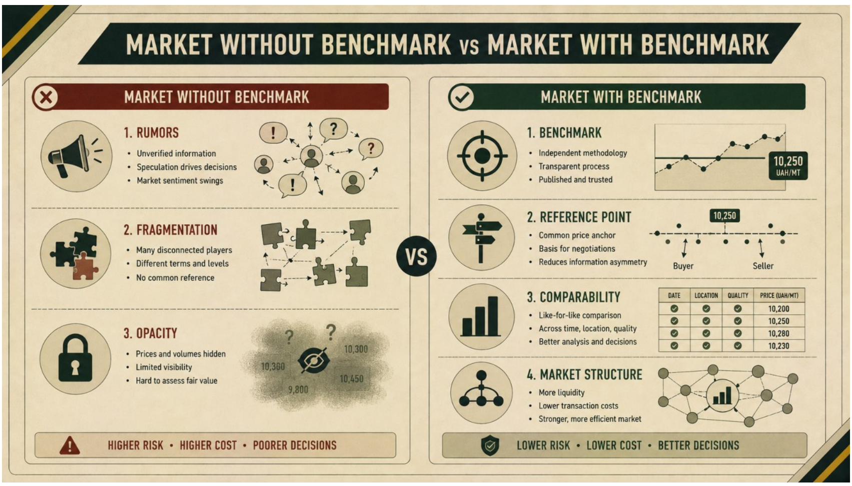
A well-designed benchmark provides that understanding without exposing individual participants. Over time, the benchmark itself becomes part of the market. Traders reference it. Analysts study it. Contracts incorporate it. Market participants use it to measure premiums, discounts and basis relationships.

The benchmark becomes a common language. And language matters. Because markets are ultimately systems of communication. Every quote is a signal. Every transaction is a signal. Every basis movement is a signal.

The challenge is not generating signals. The challenge is interpreting them.

Benchmarks help transform thousands of isolated signals into something coherent. They convert information into structure. And structure creates trust.

This explains why many of the world's most successful commodity markets eventually develop benchmark ecosystems. Not because participants need more data. But because they need a better way to organize the data they already have.



The next question naturally follows:

If benchmarks rely on many market participants rather than a single source, how do they ensure neutrality and reliability?

Chapter 6: The Respondent Model: How the Market Speaks for Itself

Every benchmark must answer a fundamental question.

Where does the price come from?

In a centralized exchange, the answer is relatively clear.

Prices emerge from orders and trades recorded in a visible market system. But many physical commodity markets do not operate this way.

They are OTC markets.

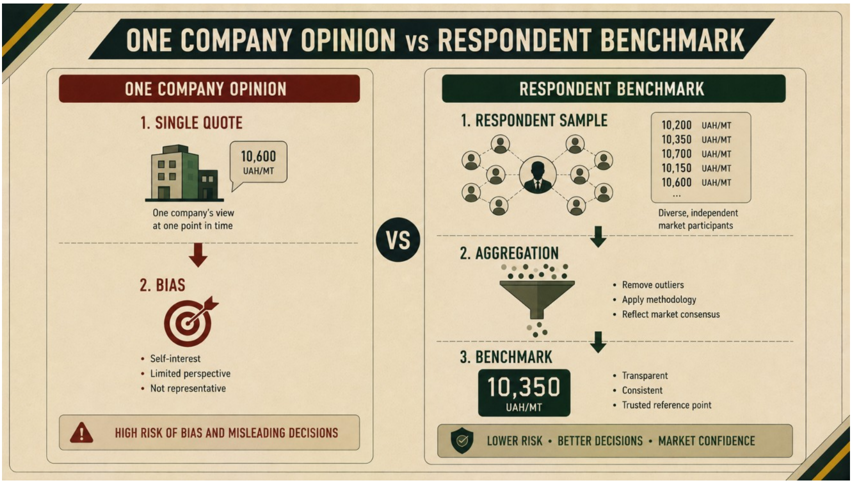
Transactions are negotiated privately. Quotes circulate through brokers, traders, processors, exporters and local buyers. Liquidity is distributed across relationships rather than concentrated in a single order book.

In such markets, a benchmark cannot simply copy an exchange price. It must be built from market participation. This is the logic behind the respondent model.

A respondent is a market participant who contributes regular price information to the benchmark process.

Respondents may include exporters, processors, traders, brokers, local buyers, elevators, producers or other active participants in the physical market. Each respondent sees part of the market. No single respondent sees everything. But together, respondents can create a more representative picture. This is the core principle of respondent-based benchmarks.

The market speaks through the people and companies who operate inside it every day. A strong benchmark should not represent one company's opinion.



Even a large and experienced participant has a limited view.

A trader may be strong in one region but weaker in another.

An exporter may see deep export flow but less local processing activity.

A processor may understand inland demand but not ocean freight.

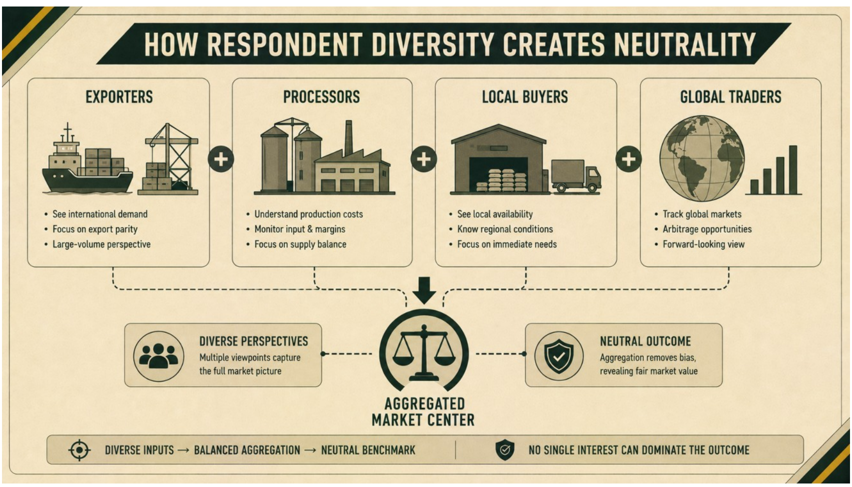
A broker may observe many conversations but not every executed transaction.

Each perspective contains value. Each perspective also contains bias. Respondent diversity helps reduce that bias.

When multiple types of participants contribute data, the benchmark becomes less dependent on any single viewpoint.

- Exporters may see one part of the market.
- Processors may see another.
- Local buyers may observe regional liquidity.
- Global trading houses may understand international demand.

Together, these perspectives help identify a market center.



This is why respondent selection matters.

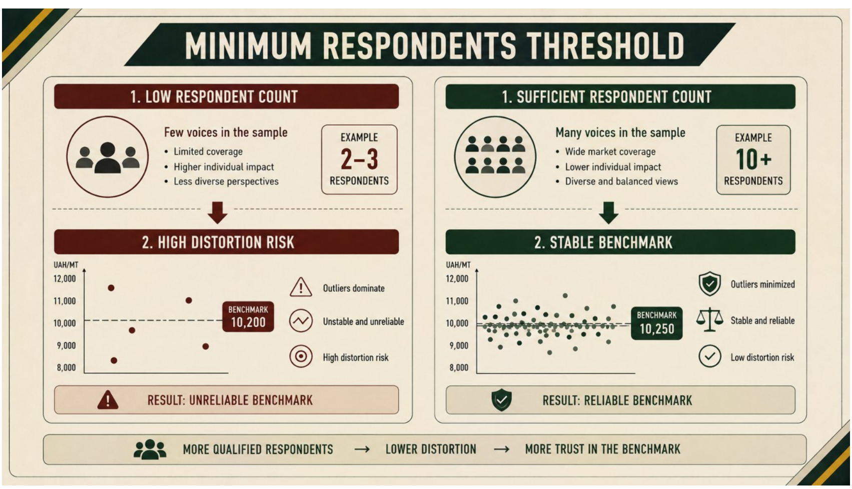
A benchmark is only as strong as the market network behind it. If the respondent base is narrow, the index becomes fragile. If it is diverse, active and representative, the benchmark becomes more credible.

Minimum respondent thresholds are therefore not administrative details. They are essential safeguards.

Publishing a benchmark based on too few inputs can create false confidence. A single unusual quote may dominate the result. A temporary local imbalance may appear to represent the market. A weak sample may be mistaken for a real market center.

For this reason, mature benchmark methodologies often require a minimum number of valid respondent inputs before publication. If the threshold is not met, the correct decision may be not to publish. This is not a failure of the benchmark. It is a sign of methodological discipline.

A benchmark should not pretend to know more than the data allows.



The respondent model also requires a careful balance between transparency and confidentiality.

Market participants may be willing to contribute data only if their individual positions remain protected. This is understandable.

Physical commodity markets are competitive.

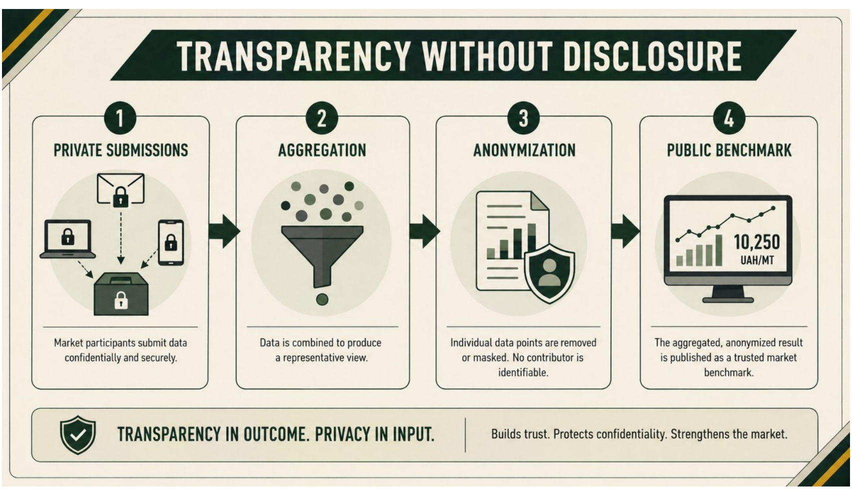
Individual bids, offers and trading levels may reveal strategy, inventory pressure, procurement needs or commercial priorities.

A benchmark must therefore protect individual submissions.

At the same time, the market needs transparency. The solution is aggregation.

Individual respondent data remains private.

The published benchmark reflects the aggregated market signal.



This allows the market to gain a public reference point without exposing confidential company-level information.

Transparency does not mean publishing every quote.

Transparency means explaining how the benchmark is built.

- *Who can participate?*
- *What data is collected?*
- *How often is it submitted?*
- *How are outliers handled?*
- *When is the benchmark published?*
- *What happens when there are insufficient respondents?*

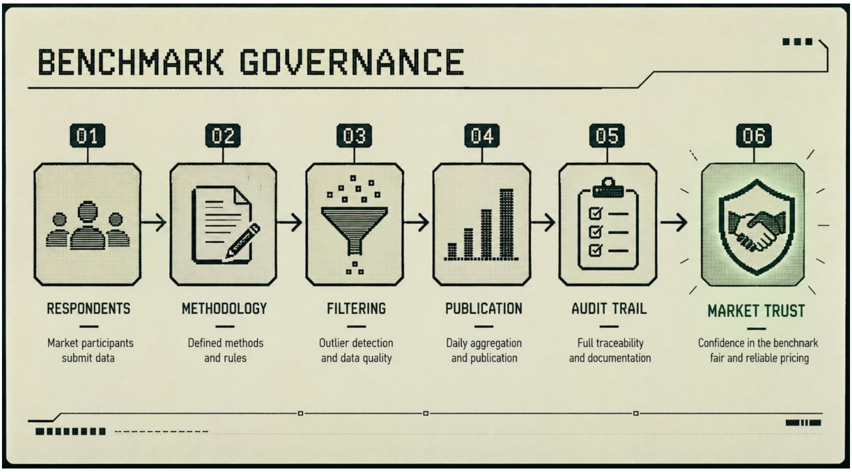
These questions define benchmark governance.

Governance is what separates a serious benchmark from a price list.

- A price list can be created quickly.
- A benchmark requires rules.
- Methodology.
- Publication discipline.
- Respondent management.
- Data validation.
- Auditability.

- Consistency.

Trust does not come from a chart. It comes from process.



Over time, respondents themselves become part of market infrastructure.

- They do not merely provide numbers.
- They help create a shared view of the market.
- They help transform private market knowledge into public market intelligence.

This is one of the most powerful aspects of respondent-based benchmarks.

The market does not wait for an external authority to define it. It begins to define itself. Through repeated participation. Through consistent methodology. Through aggregation. Through trust.

This is how fragmented physical markets begin to develop a common language.

The next challenge is methodological.

Once respondent data is collected, how should it be calculated?

And why is a simple arithmetic average often not enough?

Chapter 7: Median, Outlier Filters and the Problem of Manipulation

Once respondent data has been collected, a benchmark faces a second critical challenge.

How should that data be calculated?

At first glance, the answer appears simple: Collect the prices. Calculate the average. Publish the result.

In practice, physical commodity markets are rarely that clean: Markets contain noise. Mistakes occur. Unusual transactions happen. Liquidity can be uneven. Participants may interpret conditions differently. And occasionally, some submissions may not represent the broader market at all.

A benchmark must therefore distinguish between market information and market distortion. This is why benchmark methodology matters.

Consider a simple example.

Suppose a benchmark receives the following respondent submissions:

- 220
- 221
- 221
- 222

- 223
- 260

Five values are tightly clustered. One value is dramatically different. What should the benchmark do?

A simple arithmetic average would immediately shift upward. The extreme observation would influence the result disproportionately. The published value would move away from where most participants actually see the market.

This is the core weakness of relying exclusively on arithmetic averages. Averages are highly sensitive to outliers.

One unusual value can influence the entire calculation.

This is where the median becomes useful.

Unlike an arithmetic average, the median focuses on the center of the distribution rather than the magnitude of individual observations.

In the previous example, the median remains close to the central cluster even though one extreme value exists. This does not mean the extreme value is wrong. It simply means that one observation does not automatically redefine the market. This distinction is crucial.

Benchmarks are not trying to identify the highest quote. They are not trying to identify the lowest quote. They are trying to identify the market center.

Case 03. How Methodology Protects an Index from a Wrong Quote

Imagine a normal publication day for a Black Sea corn FOB benchmark.

Most respondent submissions arrive within a narrow range. The market appears relatively stable. Export demand is present, but not aggressive. Logistics are functioning normally. Several respondents submit values that are close to one another. Then one submission appears far away from the rest of the sample.

At first glance, this creates a problem.

The quote may be real. It may reflect a special cargo, a specific delivery window, a different quality specification, *an urgent buyer, a distressed seller or a route that is not comparable with the rest of the sample. It may also be a mistake.*

The respondent may have entered the wrong number. The quote may refer to a different delivery period. It may reflect CPT instead of FOB, or a different port, volume or execution condition.

In a weak methodology, this single observation could immediately distort the published value.

In a stronger methodology, it triggers a process.

The benchmark does not simply accept the value because it was submitted. It also does not reject it automatically. Instead, the methodology asks whether the observation represents the market center or a specific exceptional situation. This is where median logic and outlier controls become important.

The unusual value can be reviewed, checked and interpreted. If it is valid but unrepresentative, the methodology can prevent it from redefining the benchmark. If it reflects a genuine market shift confirmed by other data, the benchmark can incorporate that movement over time.

The key point is discipline.

A benchmark should not reward the most aggressive quote. It should not be pulled around by one abnormal observation. It should identify the market center using a repeatable process. This is why methodology matters as much as the number itself.

A credible benchmark is not credible because every daily value is perfect. It is credible because market participants understand how unusual observations are handled.

They know that mistakes will not automatically become market truth.

They know that one special transaction will not immediately redefine the benchmark.

They know that the published value reflects a process, not a reaction.

That is how a benchmark protects trust in a fragmented physical market.

In physical markets, market centers are often more important than market extremes.

- A single transaction may reflect a unique logistical situation.
- A distressed seller.
- An urgent buyer.
- A quality difference.
- A transportation issue.
- A regional imbalance.

Any of these factors can produce a legitimate price that is nevertheless unrepresentative of broader market conditions.

The benchmark must therefore ask a different question:

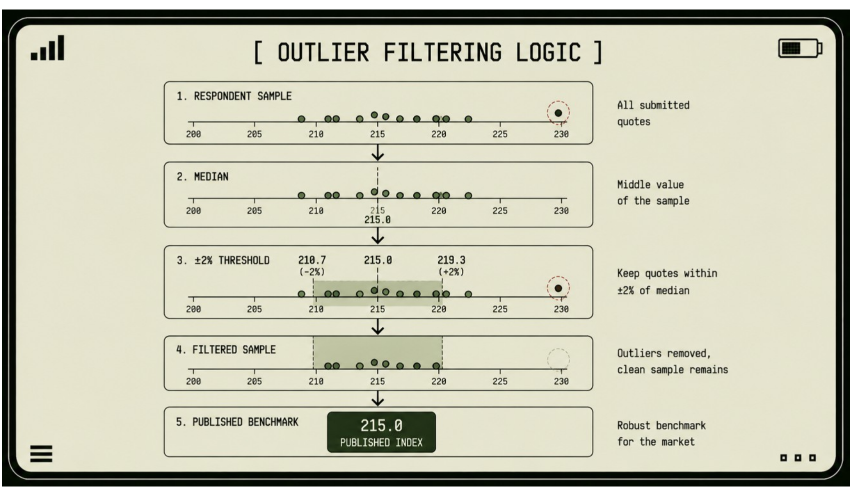
Does this observation describe the market, or does it describe a specific situation?

Outlier filtering is designed to help answer that question. An outlier filter does not claim that an observation is invalid.

It simply identifies observations that are unusually distant from the market center. These observations can then be reviewed or excluded according to the methodology.

The exact threshold may vary between benchmarks. Some use fixed percentages. Others use statistical ranges. Others combine multiple techniques.

The objective remains the same: Protect the benchmark from distortion while preserving genuine market information. This balance is important.



A benchmark that filters too aggressively may ignore legitimate market movement. A benchmark that filters too loosely may become vulnerable to noise.

Methodology therefore requires judgment. It is not enough to apply formulas mechanically. The benchmark must remain sensitive to real market change while remaining resistant to random distortions.

This challenge becomes even more important when markets are thin or fragmented. In a highly liquid market, one unusual quote is unlikely to dominate the sample. In a thin market, a single extreme value may represent a large share of all available observations. Without appropriate controls, the benchmark can become unstable. The risk is not limited to accidental errors. Markets occasionally experience intentional distortions as well.

Participants may attempt to influence perception.

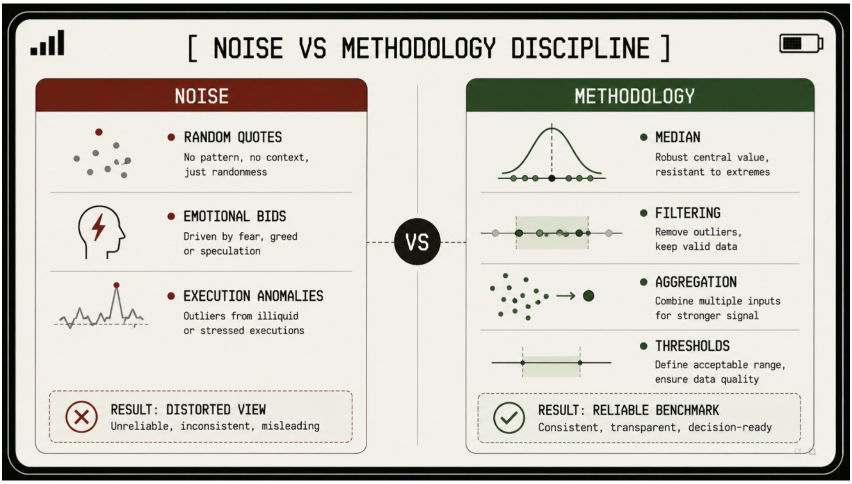
- A quote may be submitted strategically.
- An unrealistic indication may be circulated.
- An isolated transaction may be promoted as evidence of a broader trend.

This is not unique to commodity markets.

It occurs wherever information influences decisions.

A robust methodology therefore acts as a safeguard.

Median calculations. Outlier filters. Minimum respondent thresholds. Governance procedures. Auditability. Together, these mechanisms reduce the influence of isolated distortions.



Over time, they create something more valuable than a daily price.

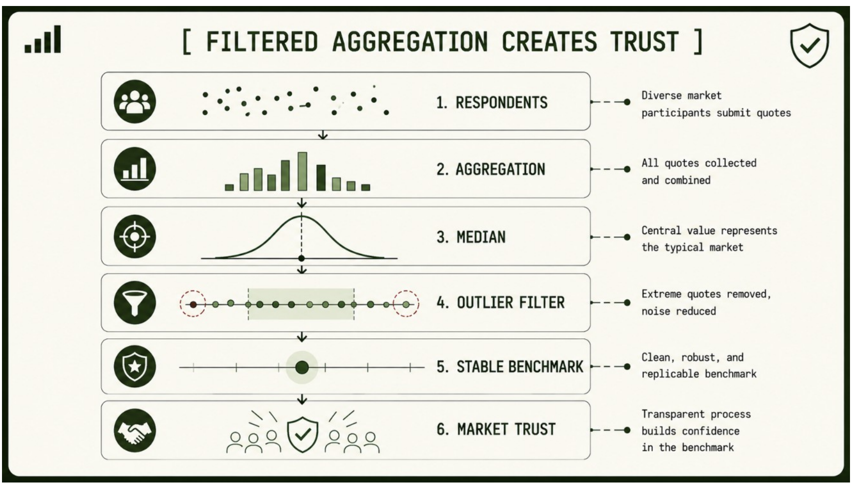
- They create trust.
- Market participants do not trust a benchmark because it always agrees with them.
- They trust it because they understand how it works.
- They know that the benchmark follows consistent rules.
- They know that unusual observations will not immediately redefine the market.
- They know that publication decisions are based on methodology rather than opinion.

- Trust is ultimately the product of repeatable process.

And in benchmark construction, methodology is that process.

The result is not perfect accuracy. Perfect accuracy is impossible in any market.

The result is something more useful: *a stable and credible representation of the market center.*



Once we understand how calculations are protected from distortion, another question emerges.

What happens when there is not enough data to calculate a reliable benchmark at all?

Chapter 8: Thin Markets, Liquidity and the Value of Missing Data

Most people assume that more published prices automatically mean better market information.

At first glance, the logic seems obvious. If a benchmark publishes a number every day, the market gains a continuous stream of information. If no number is published, information appears to be missing.

In reality, mature benchmarks often operate according to a different principle. Sometimes the most informative signal is the decision not to publish.

To understand why, we first need to understand the concept of market liquidity.

Liquidity is not simply the existence of a price. Liquidity is the ability to transact at a price.

A market can display a quote and still have very little liquidity. A market can appear active while offering few realistic execution opportunities. A market can produce a number without producing confidence. This distinction becomes especially important in thin markets.

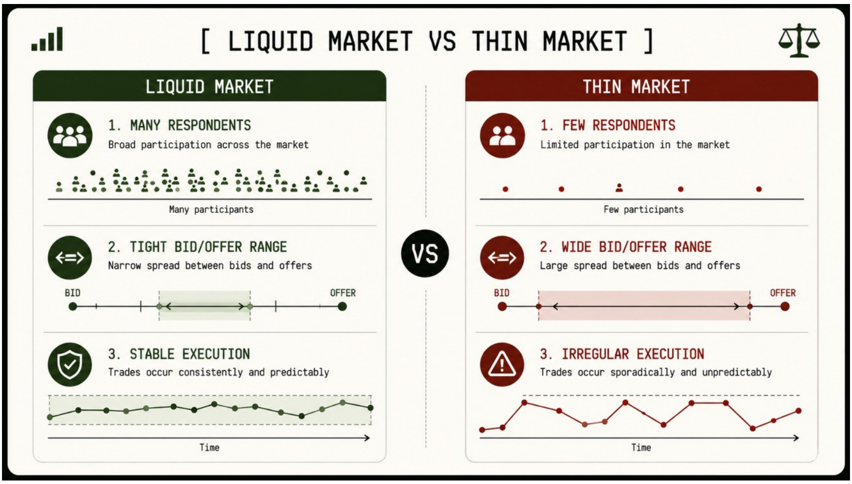
A thin market is a market with limited participation. There may be few buyers. Few sellers. Few transactions. Few respondent submissions.

Execution opportunities may appear sporadically rather than continuously.

Bid and offer ranges may be unusually wide.

In such environments, prices become less stable and less representative.

A single observation can exert a disproportionate influence on the market picture.



This creates a risk known as false precision.

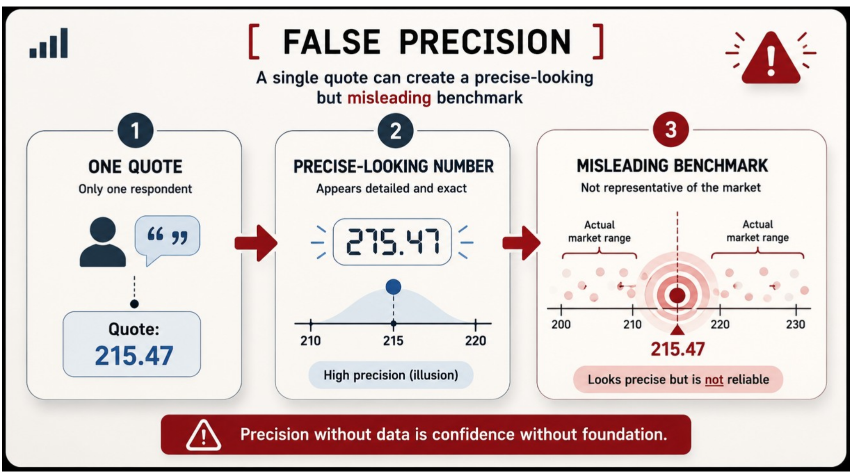
False precision occurs when a benchmark publishes a highly specific number despite having insufficient information to support it. The result appears authoritative. The underlying market reality is not. Consider a simple example.

Suppose a benchmark receives only two submissions. The values may look precise. The methodology may still produce a calculation. The benchmark may technically be capable of publishing a number. But does that number genuinely represent the market? Or does it merely represent two participants?

The answer matters.

Market participants often place considerable trust in benchmarks. If a published value appears precise, users naturally assume that it reflects broad market conditions.

When the underlying sample is weak, this assumption becomes dangerous. A benchmark may unintentionally create confidence where confidence does not exist.



This is why mature methodologies place significant emphasis on respondent thresholds and publication rules.

The goal is not to maximize publication frequency. The goal is to maximize reliability.

Sometimes this means refusing to publish.

This idea often feels counterintuitive.

Many users expect a benchmark to provide a number every day.

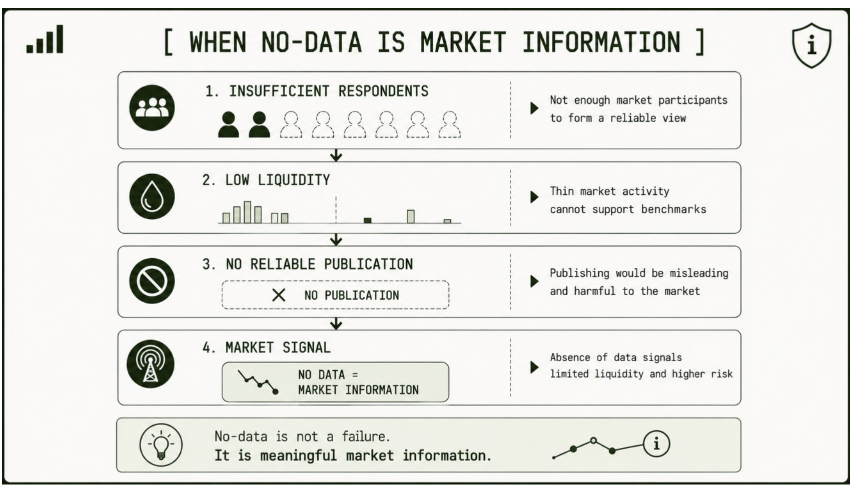
Yet a benchmark that publishes weak data may ultimately damage trust more than a benchmark that occasionally publishes nothing.

The absence of publication is not necessarily a failure. It can be information.

A missing benchmark may signal that liquidity is insufficient. It may indicate that respondent participation is too low. It may suggest that market activity is fragmented or temporarily disrupted.

In all of these cases, the absence of a benchmark conveys meaningful information about market conditions.

No-data becomes market data.



This concept is particularly important in physical commodity markets.

Unlike highly liquid financial markets, commodity markets often experience periods of uneven activity.

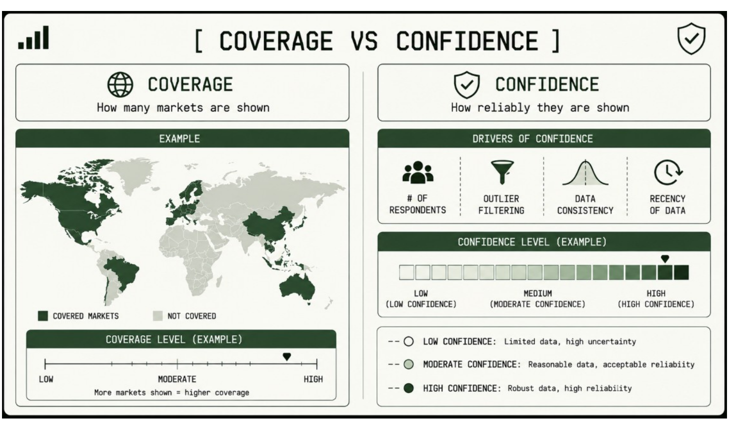
Liquidity shifts between locations. Export programs change. Regional demand fluctuates. Logistics constraints emerge and disappear. Seasonal cycles affect participation. A benchmark must recognize these realities rather than ignore them. This is why publication decisions should always balance coverage against confidence.

Coverage answers one question: *How many markets are being observed?*

Confidence answers a different question: *How reliable are the observations?*

A benchmark that maximizes coverage at the expense of confidence may publish more numbers. A benchmark that prioritizes confidence may publish fewer.

In the long run, confidence is usually more valuable.



Trust depends on confidence.

A market participant can work with incomplete information.

What they cannot work with is misleading information disguised as certainty. This is why minimum respondent thresholds are so important.

The threshold is not a bureaucratic requirement. It is a quality control mechanism. It protects the benchmark from becoming overly dependent on a small number of observations. It protects users from false precision. It preserves credibility.

The strongest benchmarks therefore share an important characteristic: *they are willing to admit uncertainty*. They are willing to acknowledge when the market does not provide enough information for reliable publication.

Case 04. Why a Benchmark May Refuse to Publish

For many users, the absence of a published benchmark looks like a problem.

The market expects a number.
Dashboards expect a number.
Reports expect a number.
Users want continuity.

If the number is missing, the natural question is: What went wrong?

In reality, the absence of publication may sometimes be more useful than publication itself. Imagine that a benchmark methodology requires a minimum number of valid respondent submissions before a daily value can be published. On a certain day, the system receives only two or three usable inputs.

Technically, a calculation is possible.

The system could average the available submissions. It could produce a clean number. It could even publish the value with decimal precision. But would that number represent the market? Probably not.

With such a small sample, each individual submission carries too much weight. One unusual quote, one specific logistics case, one local imbalance or one participant's commercial strategy could dominate the result.

The benchmark would create the appearance of precision without the substance of market confidence. This is false precision. A mature benchmark must avoid it.

The correct decision may therefore be not to publish the value at all, or to mark the basket as insufficiently liquid.

At first glance, this may look less informative.

In reality, it is a sign of methodological discipline.

The benchmark is telling the market: There is not enough reliable information today to form a credible market center. This message has value.

It tells traders that liquidity is weak. It tells analysts that the market is fragmented. It tells risk managers that the published history should not be filled with artificial numbers. It tells users that when the benchmark does publish, the value is supported by sufficient evidence.

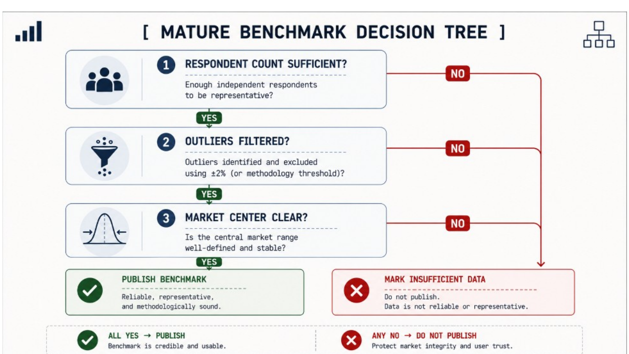
Paradoxically, this willingness often increases trust.

Market participants recognize that the benchmark values integrity over appearance.

Over time, they learn that a published value means something. It represents not merely a calculation, but a market condition supported by sufficient evidence.

This is one of the most important distinctions between a benchmark and a simple price list. A price list publishes values whenever values exist. A benchmark publishes values when confidence exists. The difference may appear subtle.

In practice, it is foundational. Because benchmark quality depends not only on how prices are calculated. It also depends on knowing when not to calculate them.



Once we understand liquidity and publication confidence, another challenge emerges.

Even when sufficient data exists, local physical markets often behave differently from global benchmarks.

This brings us to one of the most important concepts in commodity trading: *basis risk*.

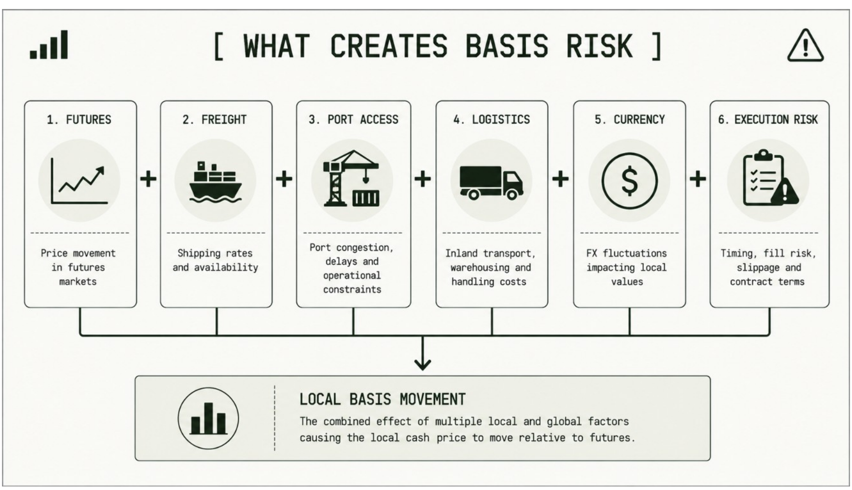
Chapter 9: Basis Risk: Why Local Prices Can Move Separately from Global Benchmarks

Basis risk is one of the most important concepts in physical commodity markets. It is also one of the most misunderstood.

Many people outside the market assume that if a trader follows a global futures benchmark, they already understand the price.

In reality, futures prices explain only part of the story. The physical market adds another layer. That layer is basis.

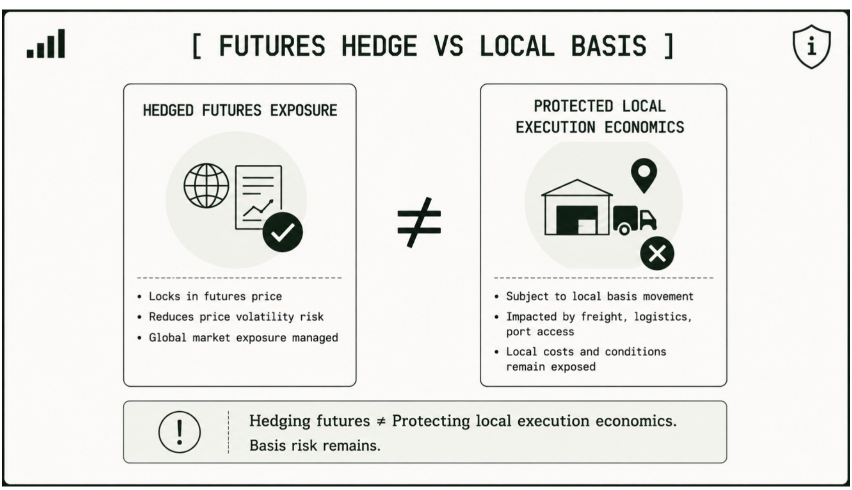
Basis is the difference between a reference price and a local physical market value.



It reflects everything that makes one market different from another:

- location;
- transportation;
- freight;
- storage;
- quality;
- currency;
- infrastructure;
- liquidity;
- execution risk;
- local supply and demand.

In simple terms, basis is where global price expectations meet local reality.



This is why basis can move even when futures prices do not.

- A futures market may remain stable while a local basis weakens.
- A futures market may rise while a physical export price falls.
- A futures market may decline while a local market strengthens because local buyers urgently need supply.

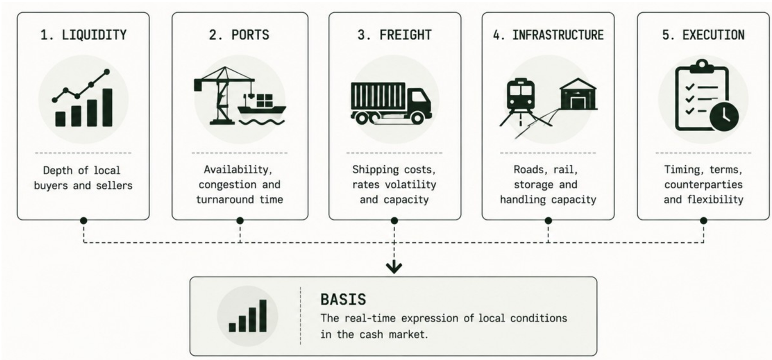
These situations can surprise those who view commodity prices only through the lens of global benchmarks. But they are normal in physical markets.

A futures price represents a standardized contract. It is designed to concentrate liquidity and help participants manage price risk.

A local physical price represents the actual economics of buying, selling, moving and executing a commodity in a specific place.

The two are related. But they are not interchangeable. This distinction matters for hedging.

A futures hedge can reduce exposure to broad price movement. But it does not eliminate all local market risk.



Suppose a trader hedges futures exposure correctly. The global price risk is partially controlled. But then freight costs rise. A port becomes congested. Local demand weakens. Currency conditions change. The basis moves against the trader. The futures hedge may work exactly as intended, yet the physical margin may still deteriorate. This is basis risk. It is the residual risk that remains when local physical market conditions change differently from the global reference.

Basis risk is not a flaw in commodity markets. It is a natural feature of them.

Commodities are physical goods. They move through imperfect infrastructure. They depend on routes, terminals, vessels, trucks, railways, warehouses, inspection systems and financing conditions.

Every local market has its own constraints. Every constraint affects price. This is why basis is often described as the language of the physical market.

It tells us what futures prices cannot fully show.

- *Is the local market strong or weak?*
- *Is logistics becoming expensive?*
- *Are buyers active?*
- *Is export demand present?*
- *Is liquidity concentrated in one route?*
- *Is execution risk increasing?*
- *Is local supply under pressure?*

Basis answers these questions indirectly. It transforms local conditions into price relationships. For this reason, basis intelligence is essential for serious commodity analysis.

Global futures provide the macro signal. Local spot indices provide the physical signal. Together, they allow market participants to understand the full price structure.

Without local basis data, market analysis remains incomplete.

This is especially true in regions where logistics are complex, export routes are unstable or physical liquidity is fragmented. In such environments, global benchmarks alone can be misleading.

A rising futures market may create the impression of strength. But if local logistics deteriorate, physical prices may not follow.

A falling futures market may appear bearish. But if local buyers urgently need supply, local basis may strengthen.

The outright price may tell one story.

Basis may tell another. This is why spot indices become valuable. They provide a structured way to observe local physical prices. When combined with futures data, logistics data and historical spreads, they create a basis intelligence layer. This layer helps traders and analysts understand not only where the market is, but why it is there.

Case 05. Chicago Rises while FOB Black Sea Weakens

Imagine the following situation. At the beginning of the week, the global market receives news about deteriorating weather conditions in the United States. Analysts begin to revise corn crop expectations. Speculative funds increase long exposure. Chicago Corn Futures rise for several days.

If we look only at the futures screen, the conclusion seems obvious: The grain market is moving higher. But at the same time, something different is happening in the Black Sea physical market.

Ports are congested. Freight costs are rising. Exporters already hold significant inventories and are not in a hurry to increase procurement prices. Some buyers reduce activity because export programs are uncertain. Execution windows become tighter. Local seller pressure remains high.

As a result, FOB Black Sea does not follow Chicago.

It may remain flat.

It may even weaken.

For an outside observer, this looks contradictory. How can futures rise while the physical market falls?

There is no contradiction.

Chicago reflects global expectations.

FOB Black Sea reflects local physical execution.

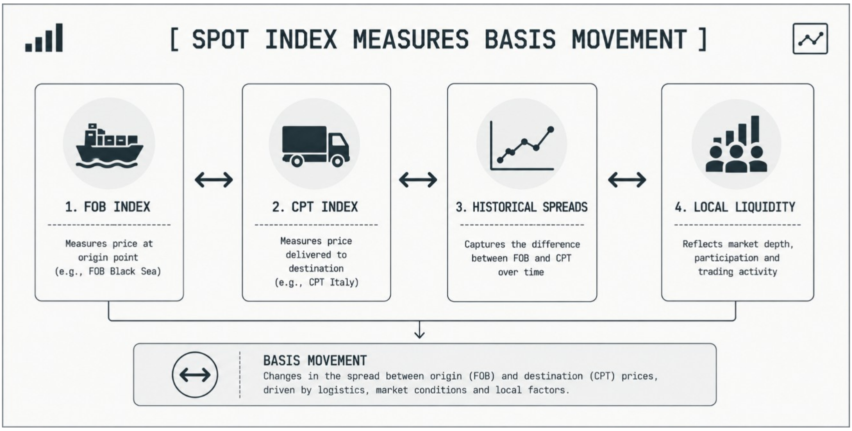
The difference between these two movements is basis risk.

For a trader, this distinction is not academic. It can directly affect margin. A futures hedge may protect part of the global price exposure, but it cannot fully protect against local logistics deterioration, weaker export demand, rising freight, port congestion or widening FOB discounts.

This is why professional market participants monitor more than futures. They watch whether FOB values, CPT values, route spreads, liquidity and basis are confirming or rejecting the global futures signal.

A rising futures market may create the impression of strength. But if local execution becomes more difficult, the physical price may not follow. In some cases, it may move in the opposite direction. This is why basis intelligence is essential in export-oriented agricultural markets.

It helps the market understand not only whether global prices are rising or falling, but whether local physical reality is confirming or rejecting that movement.



It becomes possible to monitor:

- FOB values;
- CPT values;
- farmgate prices;
- processing bases;
- freight relationships;
- route spreads;
- regional liquidity;
- historical basis movement.

Over time, this information becomes part of daily market intelligence.

The market no longer relies only on impressions.

It can observe how local physical reality behaves relative to global benchmarks.

- Basis risk cannot be eliminated.
- But it can be measured more clearly.
- It can be discussed more precisely.
- It can be managed more intelligently.

That is one of the central functions of local spot indices.

- They do not replace futures markets.
- They complete the picture.
- They show what happens when global expectations meet physical execution.
- And in commodity trading, that meeting point is often where the real market begins.

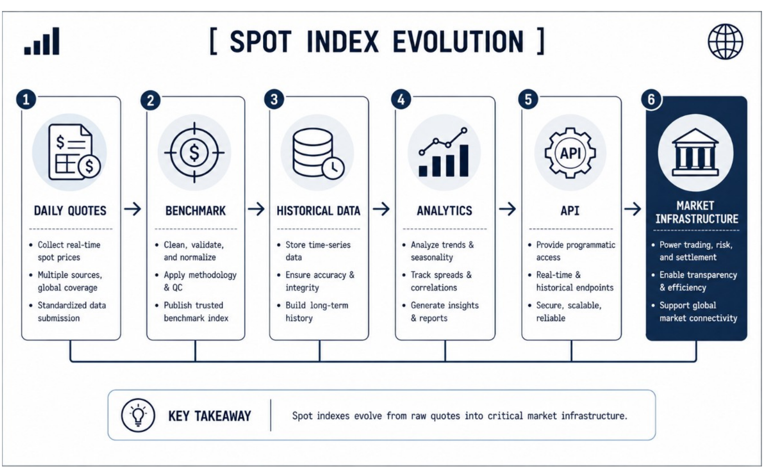
Chapter 10: Spot Indices as Market Infrastructure

Most market participants initially view an index as a number. A chart. A publication. A daily benchmark. Something useful for reference. This is understandable.

A benchmark often enters the market as a simple information product. It publishes values. It provides visibility. It helps participants understand where the market is today. But mature benchmarks rarely remain simple publications. Over time, they evolve. They become infrastructure. This transformation is one of the most important developments in modern commodity markets.

The process is gradual. At first, a benchmark helps market participants understand prices. Then it begins to help them compare prices. Later, it becomes a tool for analysis.

Eventually, it becomes part of contracts, workflows, risk systems and market communication.



At that point, the benchmark is no longer merely describing the market. It is helping the market function.

This is what market infrastructure means. Infrastructure is not necessarily visible. Participants often notice it only when it is missing. Ports are infrastructure. Communication networks are infrastructure. Payment systems are infrastructure. Benchmarks can become infrastructure as well. Because markets need common reference points.

Without a shared benchmark, participants spend significant effort debating basic questions.

- *Where is the market?*
- *Which quote should be trusted?*
- *Which transaction is representative?*
- *Which region should define the reference price?*

Every participant may have a different answer. A benchmark does not eliminate disagreement. But it creates a common frame of reference. The market gains a shared language.

Once that language exists, new possibilities emerge.

- Premiums can be measured consistently.
- Discounts can be compared.
- Basis relationships become easier to analyze.
- Regional markets become easier to evaluate.
- Contracts become easier to structure.

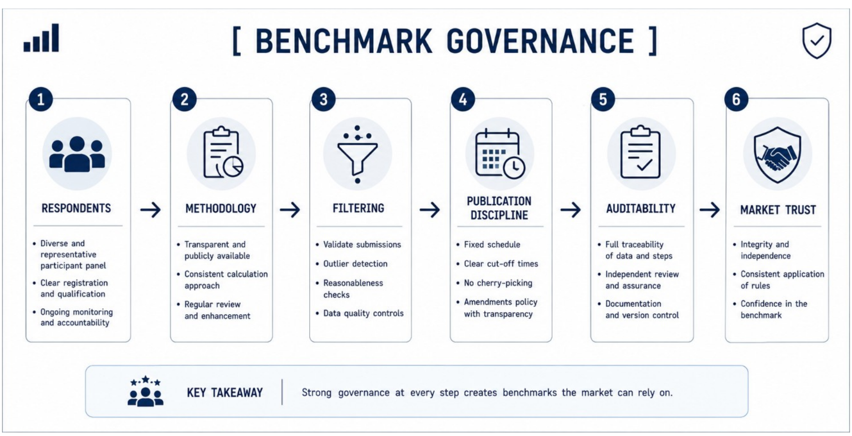
This is why benchmark governance matters so much.

A benchmark cannot become infrastructure unless participants trust it. Trust does not emerge automatically.

It is created through process.

- Respondent selection.
- Methodology.
- Filtering.
- Publication discipline.
- Auditability.
- Transparency.
- Consistency.

These elements are not administrative details. They are the foundation of credibility.



A benchmark with weak governance remains a publication. A benchmark with strong governance can become infrastructure. This distinction becomes especially important when benchmarks begin to enter commercial agreements.

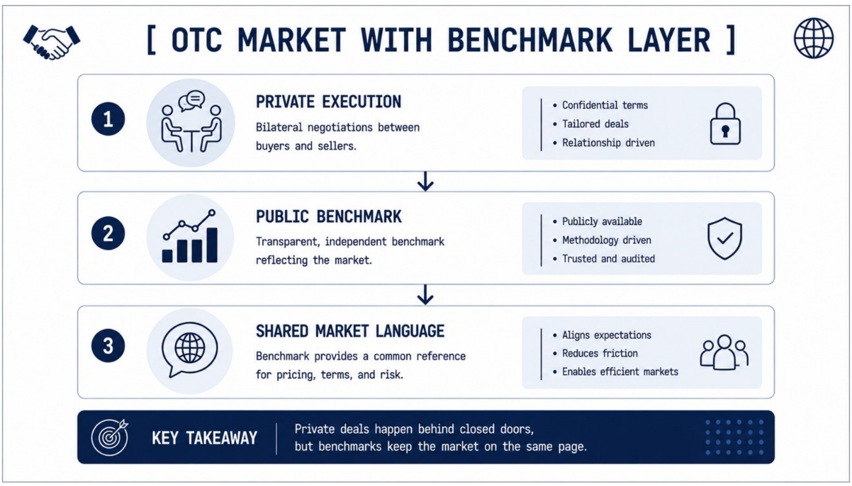
Many commodity contracts eventually reference benchmark values. Pricing formulas become possible. Premiums and discounts can be calculated relative to a common index. Negotiations become more efficient because both parties share a reference point. The benchmark reduces friction.

Instead of debating where the market is, participants can focus on how their specific situation differs from the market.

This function is valuable even in OTC markets. Perhaps especially in OTC markets.

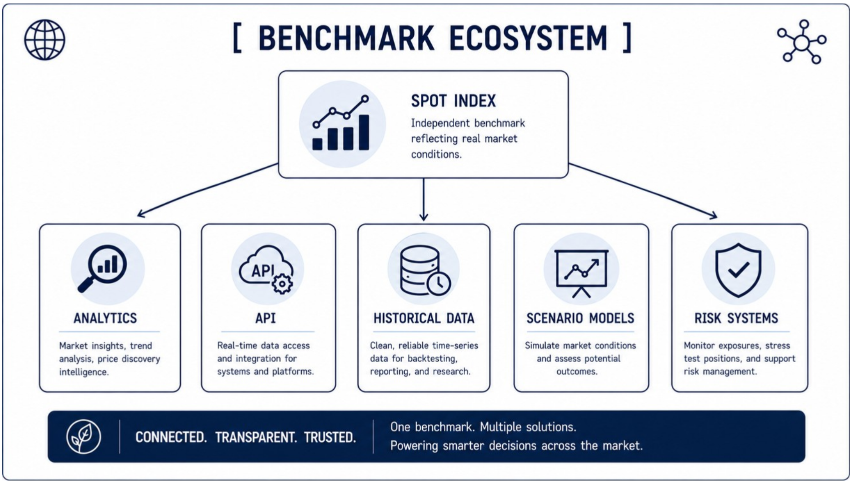
Physical commodity markets often rely on private execution. Transactions remain confidential. Relationships remain important. Execution remains decentralized.

Yet even decentralized markets benefit from a public reference layer. A benchmark provides exactly that. Private execution remains private. Market understanding becomes public. This combination is powerful. It allows markets to maintain flexibility while improving transparency.



Over time, the benchmark begins to attract additional layers. Historical datasets emerge. Analytics products appear. Market reports become more sophisticated. Spreads and basis relationships can be monitored systematically. APIs can distribute data automatically. Researchers and analysts gain structured information.

The benchmark evolves into an ecosystem.



This is where projects such as **1D3X** begin to differ from traditional market publications. The objective is not merely to publish daily values.

The objective is to create infrastructure that can support:

- market intelligence;
- analytics;
- contracts;
- risk management;
- data products;
- future trading systems.

An index is therefore not the end product. It is the foundation layer.

Everything else is built on top of it. This perspective changes how we think about commodity markets. Instead of viewing benchmarks as reports, we begin to view them as networks.

Networks connecting respondents, traders, processors, exporters, analysts, institutions and technology platforms.

The benchmark becomes the center of that network. Information flows through it. Market participants coordinate around it. New services emerge from it.

This is why modern benchmark projects increasingly focus on ecosystems rather than publications. The benchmark itself remains important. But its greatest value may lie in what it enables.

A single daily number is useful. A trusted market infrastructure layer is transformational. And once a benchmark becomes infrastructure, another opportunity appears.

The data itself becomes reusable.

- It can power analytics.
- It can support automation.
- It can feed APIs.
- It can help build market intelligence systems.

This is where commodity markets begin their next stage of evolution.

Chapter 11: Daily Market Intelligence: Why Structured Data Matters

Markets generate information continuously.

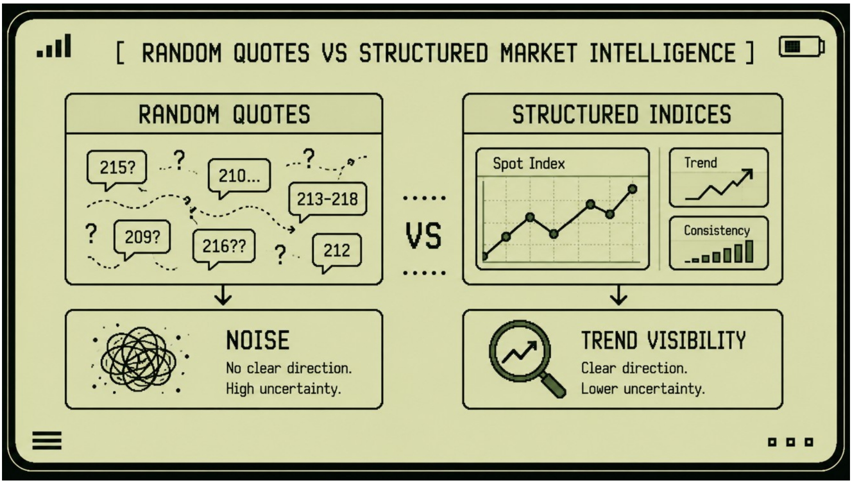
- Every conversation.
- Every quote.
- Every bid.
- Every offer.
- Every shipment.
- Every execution.
- Every logistical disruption.
- Every currency move.
- Every weather event.

Each of these contributes to market understanding.

Yet information alone does not create intelligence.

In many commodity markets, participants have access to enormous amounts of information but relatively little structure. This distinction is important.

Information answers isolated questions. Intelligence explains relationships.



A trader may know today's price. Intelligence helps explain whether that price matters. A broker may observe a bid. Intelligence helps determine whether the bid reflects a broader trend.

A market participant may notice volatility. Intelligence helps distinguish between temporary noise and structural change. This is where structured data becomes valuable.

Structured data transforms observations into context.And context changes decisions.

Consider the difference between seeing a single price and seeing a historical series of prices.

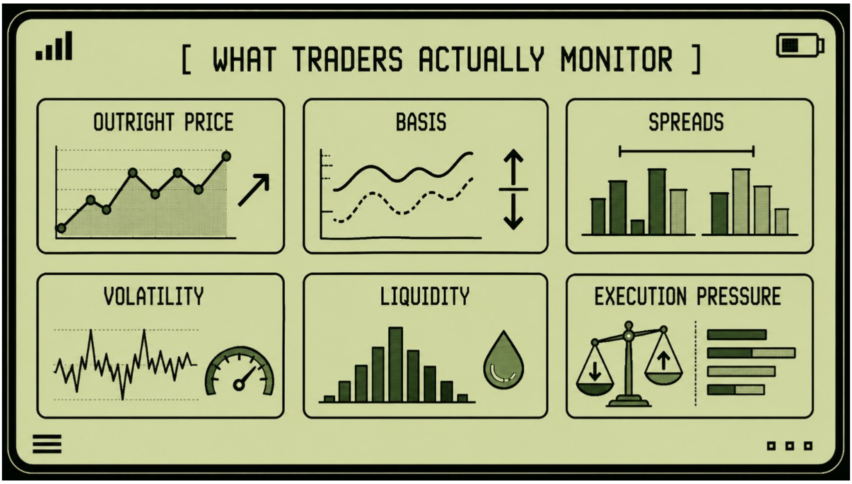
The first tells you where the market is today. The second tells you how the market arrived there. The distinction may seem subtle.

In practice, it changes everything. Commodity markets rarely move in isolation. Prices interact with logistics. Basis interacts with liquidity. Spreads respond to infrastructure. Freight affects competitiveness. Regional markets influence one another. Without historical continuity, these relationships are difficult to see. With historical continuity, patterns begin to emerge.

This is why professional traders rarely monitor price alone. They monitor structure. Price is only one variable among many.

A typical commodity trader may simultaneously observe:

- outright prices;
- basis relationships;
- regional spreads;
- liquidity conditions;
- freight costs;
- currency movement;
- execution pressure;
- market volatility.



Taken individually, these signals can appear disconnected. Together, they form a picture.

The challenge is organizing them. A benchmark helps because it creates consistency. A daily spot index generates a repeatable observation. Each publication becomes a reference point.

Over time, these reference points accumulate. A historical dataset emerges.

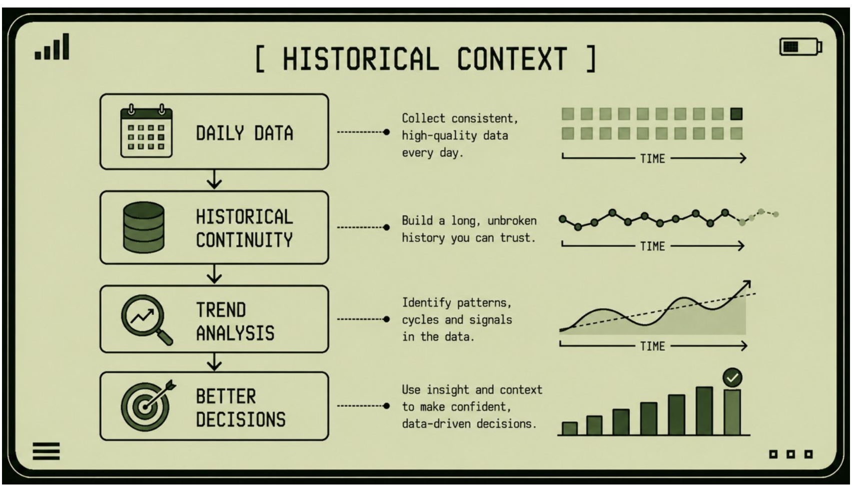
Once a dataset exists, market participants gain something they previously lacked: *memory*.

Markets often suffer from short-term attention. Participants focus on today's news. Today's disruption. Today's quote. Historical data creates perspective.

It allows traders to compare current conditions with previous periods.

- *Has the basis widened before?*
- *Is this volatility unusual?*
- *Have freight spreads reached similar levels in the past?*
- *Is local liquidity behaving differently from historical norms?*

Without structured history, these questions become difficult to answer. With structured history, they become measurable.



This is one reason spot indices are increasingly valuable.

They do not simply record prices. They record market evolution.

As datasets grow, additional forms of intelligence become possible.

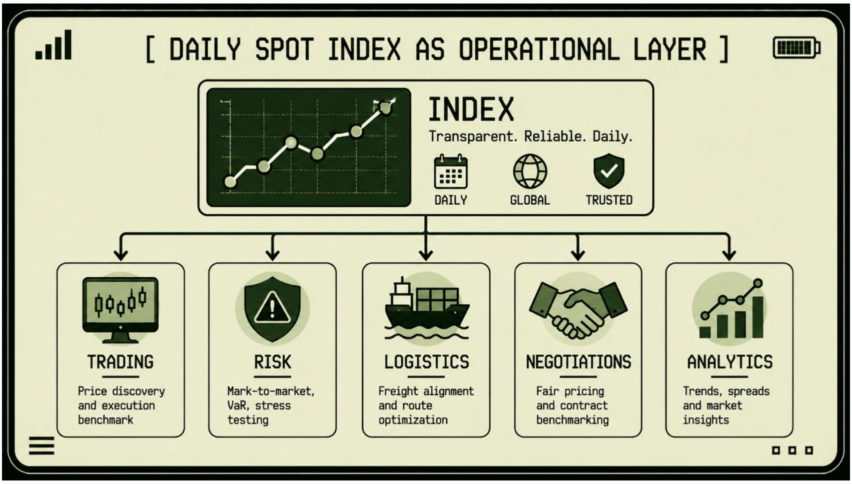
- Spread analysis.
- Volatility monitoring.
- Basis tracking.
- Seasonality studies.
- Liquidity analysis.
- Correlation research.
- Scenario modeling.

The benchmark becomes more than a publication. It becomes a daily operational layer.

Participants begin to use it as part of their workflow.

- Morning market reviews.
- Risk assessments.
- Procurement decisions.
- Export planning.
- Negotiation preparation.
- Execution monitoring.

The benchmark becomes integrated into decision-making. This is a significant shift.



Many markets begin by treating benchmarks as reports. Over time, benchmarks become operational tools. Participants no longer consult them occasionally. They rely on them continuously. This transition also changes how markets think about analytics.

Traditional analysis often focuses on explaining what happened. Modern market intelligence increasingly focuses on understanding what might happen next. This does not mean predicting the future perfectly.

Commodity markets remain uncertain. Unexpected events will always occur.

But structured data allows participants to evaluate scenarios more systematically.

- *What happens if freight increases?*

- *What happens if liquidity shifts?*
- *What happens if export demand weakens?*
- *What happens if basis diverges from futures?*

The goal is not certainty. The goal is improved decision quality.

Structured intelligence reduces dependence on intuition alone. It complements experience with evidence. This does not replace human judgment.

Commodity markets remain deeply human. Relationships matter. Negotiation matters. Execution matters. Local expertise matters. But structured data enhances these capabilities. It helps participants understand the environment in which decisions are made.

As a result, modern commodity markets are becoming increasingly data-driven. Not because people matter less. But because complexity requires better tools.

Spot indices are one of those tools. They provide continuity where markets provide fragments. They provide memory where markets provide events. They provide structure where markets provide signals.

And they create the foundation for something even larger: *market intelligence systems that can operate continuously rather than episodically.*

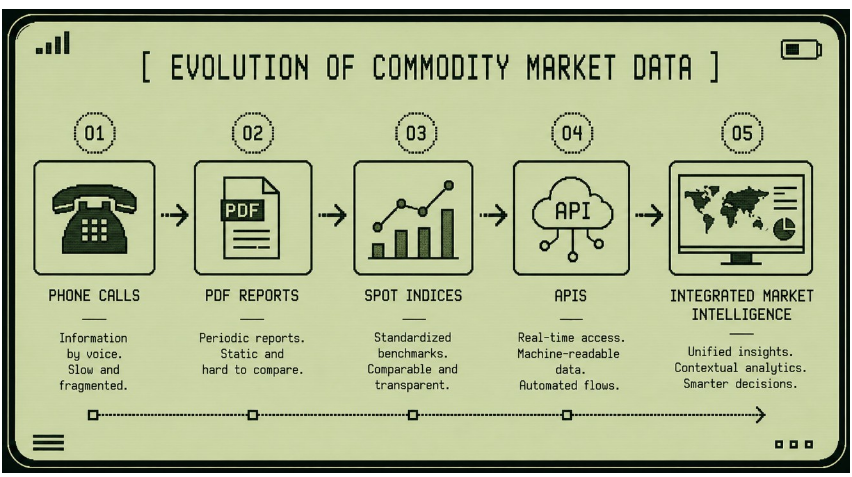
This evolution leads naturally to the next question: *If commodity markets are becoming increasingly data-driven, what happens when benchmark data becomes machine-readable and programmable?*

Chapter 12: From Index to API: The Future of Commodity Data

The history of commodity market information is, in many ways, a history of formats.

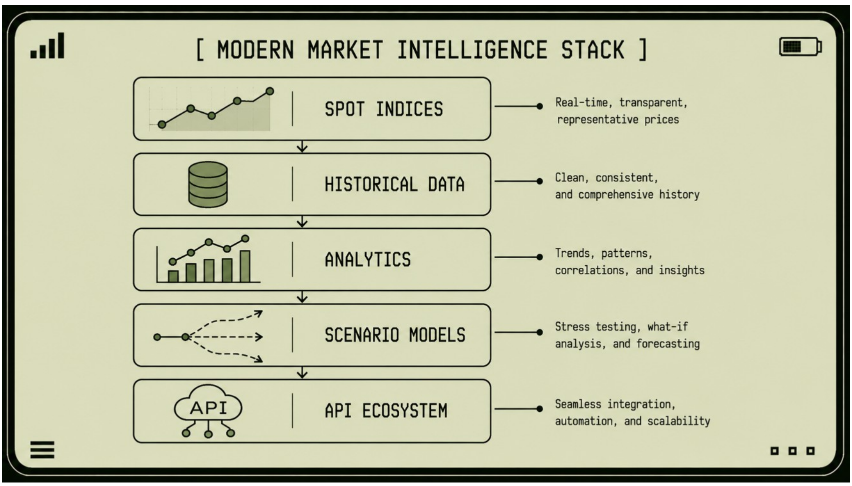
First, markets relied on conversations. Phone calls. Private meetings. Broker messages. Trading desk notes. Then came spreadsheets. Reports. PDFs. Market newsletters. Dashboards. Each format improved visibility. But each format also had limits.

A phone call cannot scale. A spreadsheet can become outdated. A PDF is difficult to integrate. A dashboard is useful for humans, but not always useful for systems.



Modern commodity markets increasingly require something more.

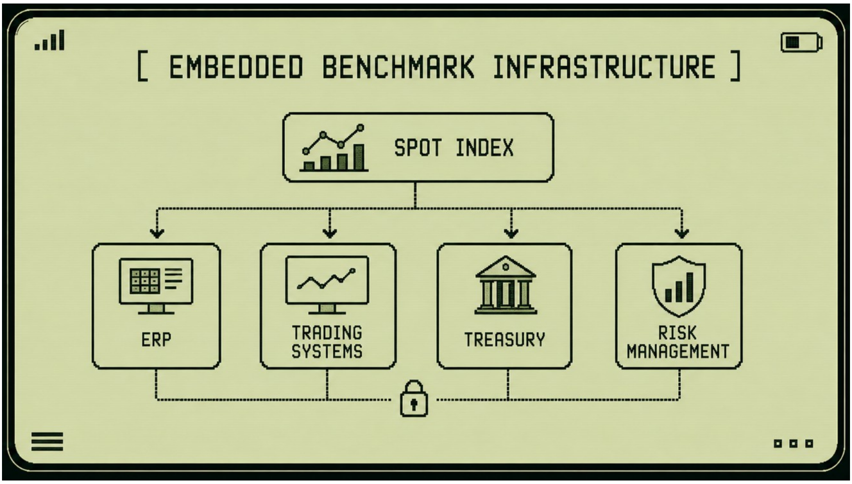
They need data that can move. Data that can be integrated. Data that can be read by machines. Data that can support decisions automatically. This is where the API becomes important.



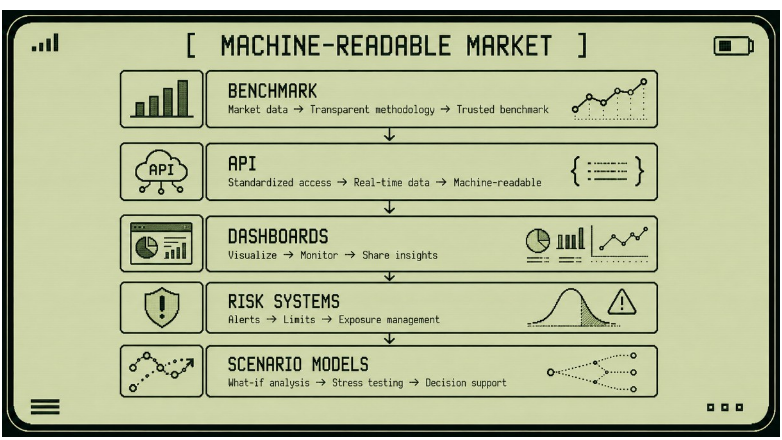
An API is not merely a technical feature. It is a new form of market access.

When benchmark data becomes available through an API, it can move directly into the systems where decisions are made. Trading systems. Risk systems. Treasury tools. Procurement dashboards. Analytics platforms. Contract management software. AI-assisted research environments. The benchmark no longer sits on a website waiting to be viewed.

It becomes embedded in workflows.



This changes the role of market data. A published index tells users what happened. A machine-readable index can help systems react, compare, monitor and model continuously. This does not mean replacing people. It means connecting market intelligence to the tools people already use.



Consider a procurement team monitoring several regional commodity markets.

Without structured data, the team may rely on manual updates, broker messages and internal spreadsheets. With API-connected benchmarks, the same team can monitor local price movements automatically.

They can compare regions. Track basis changes. Set alerts. Identify abnormal spreads. Measure volatility. Review scenarios. The work becomes faster and more structured. The same applies to traders.

A trader does not need an API because they cannot read a chart. A trader needs an API because their workflow increasingly depends on systems. Pricing models. Position monitoring. Risk limits. Scenario engines. Historical comparison tools. Decision support.

APIs allow benchmark data to become part of these systems. This is why the future of commodity data is not simply more data. It is better-connected data.

A market does not become intelligent because it produces thousands of numbers. It becomes intelligent when those numbers can be organized, interpreted and used.

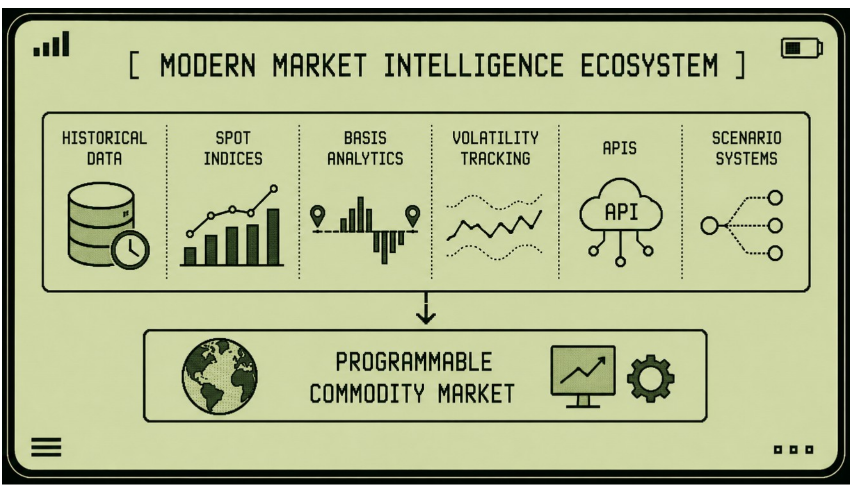
This is the logic behind the evolution from index to API.

First, a benchmark creates a trusted reference point. Then daily publications create historical continuity.

Historical continuity enables analytics. Analytics enables decision support. APIs allow the entire system to become integrated.

Once this happens, benchmark infrastructure can support more advanced capabilities.

- Scenario modeling.
- AI-assisted analysis.
- Predictive indicators.
- Automated alerts.
- Contract-linked pricing.
- Risk dashboards.
- Market stress signals.
- Programmable trading infrastructure.



Each of these depends on one foundational requirement: *structured, reliable and methodologically consistent data.*

This is why benchmark governance remains important even in highly digital environments.

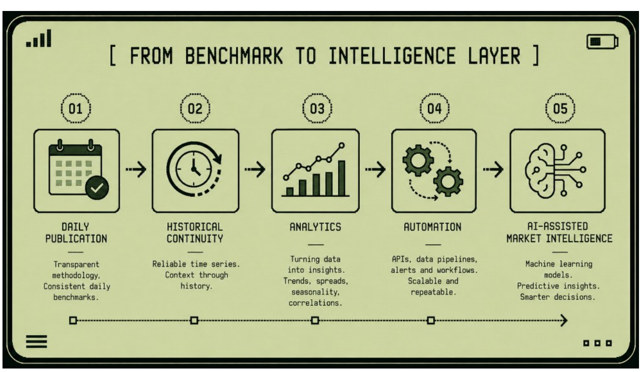
AI cannot fix poor data. Automation cannot create trust where methodology is weak. APIs do not transform noise into intelligence by themselves. The quality of the system still depends on the quality of the benchmark.

This is a central principle behind **1D3X**, **MN7R** and **Cropto**.

- **1D3X** focuses on building local commodity index infrastructure.
- **MN7R** focuses on intelligence, analytics and operational decision support.
- **Cropto** explores how benchmark data may eventually become part of programmable and tokenized commodity trading models.

These layers are connected.

A reliable index creates data. Data creates intelligence. Intelligence enables infrastructure. Infrastructure enables new market models.



This does not mean that commodity markets will become fully automated.

- Physical markets will remain physical.
- Cargoes must still move.
- Contracts must still be executed.
- Quality must still be inspected.
- Logistics must still be managed.
- Relationships will still matter.

But the informational layer around these markets is changing.

Commodity markets are becoming more measurable.

More connected.

More structured.

More programmable.

And this transformation begins not with speculation or technology hype, but with something much more basic: *a reliable benchmark.*

Because before a market can become digital, it must become readable. Before it can become programmable, it must become structured. Before it can become investable at scale, it must create trusted reference points. This is the future of commodity data. Not data as decoration.

Data as infrastructure.

Conclusion: From Market Data to Market Infrastructure

Commodity markets are becoming more complex. Not less.

- More data is available than ever before.
- More reports.
- More dashboards.
- More analytics.
- More signals.
- More noise.

Paradoxically, this abundance of information often makes markets harder to understand rather than easier.

The challenge facing modern commodity markets is therefore not information scarcity. It is information organization.

The markets that perform best are not necessarily those that generate the most data. They are the markets that create the most useful structure around that data.

This is where benchmarks become important.

- At first, a benchmark appears simple.
- A daily value.
- A chart.
- A publication.

- A reference point.

Over time, however, the benchmark begins to serve a larger purpose.

- It helps markets coordinate.
- It creates comparability.
- It supports contracts.
- It enables analytics.
- It provides historical continuity.
- It helps participants discuss the market using a shared language.

Eventually, the benchmark becomes part of infrastructure. This evolution is visible across many industries.

Financial markets rely on reference rates. Energy markets rely on benchmark pricing systems. Freight markets rely on shipping indices. Commodity markets increasingly rely on local benchmark networks. Agricultural markets are following the same path. This transformation does not eliminate market complexity.

A transparent market is not necessarily a simple market: Basis risk remains. Logistics remain. Liquidity challenges remain. Execution uncertainty remains. Physical realities remain.

What changes is the ability to observe these realities.

A good benchmark does not simplify the market by removing complexity. It simplifies understanding by organizing complexity.

This distinction is important.

Markets do not need artificial certainty. They need better visibility. This is one reason respondent-based benchmarks have become increasingly valuable. They allow markets to describe themselves. Not through isolated opinions. Not through individual transactions. But through structured aggregation and transparent methodology.

The result is not perfect accuracy. No benchmark can perfectly represent every participant, every transaction or every local condition.

The result is something more practical: *a trusted approximation of the market center*.

- Trust is ultimately the key word.
- Trust in methodology.
- Trust in governance.
- Trust in publication discipline.
- Trust in transparency.
- Trust in market infrastructure.

Without trust, a benchmark is merely a number. With trust, it becomes part of the market itself.

This idea sits at the center of projects such as **1D3X**, **MN7R** and **Cr0pto**.

Each project addresses a different layer of the same challenge.

- **1D3X** focuses on building benchmark infrastructure for local commodity markets.
- **MN7R** focuses on transforming market observations into usable intelligence.
- **Cr0pto** explores how benchmark-driven markets may eventually become programmable, tokenized and globally connected.

Together, these initiatives reflect a broader belief: *the future of commodity markets is not only physical.*

It is informational.

- Physical flows will always matter.
- Crops must still be grown.
- Cargoes must still move.
- Contracts must still be executed.

But around these physical processes, a new layer is emerging.

- Indices.
- Historical datasets.
- APIs.
- Analytics.

- AI-assisted intelligence.
- Digital market infrastructure.

These tools do not replace the market.

They help the market understand itself.

And perhaps that is the most important role of a benchmark.

- Not to predict the future.
- Not to eliminate uncertainty.
- Not to simplify reality.

But to make reality more readable. Because the markets that thrive in the future will not necessarily be the ones with the most information. They will be the ones that understand their information best.

Spot indices are only the beginning of that transformation.

The larger journey is the evolution from fragmented signals to shared intelligence. From isolated observations to market memory. From market data to market infrastructure. And that journey has only just begun.

Glossary of Commodity Market Terms

Basis

Basis is the difference between a reference market price and a local physical market price.

In agricultural commodity markets, basis acts as a bridge between global price discovery and local market reality. A futures contract may reflect worldwide expectations about supply and demand, while basis reflects transportation costs, logistics conditions, local liquidity, storage availability, regional competition and execution risk.

Basis explains why local prices can strengthen while futures prices weaken, or why futures prices can rally while local export values remain under pressure. For physical traders, basis is often as important as the outright commodity price itself because it reveals what is happening inside the local market.

Benchmark

A benchmark is a structured market reference designed to represent the market center at a specific moment in time.

A benchmark is not a forecast, an opinion or a guaranteed transaction price. Instead, it is a transparent and repeatable representation of market conditions based on a defined methodology.

Benchmarks allow traders, processors, exporters, analysts and investors to discuss market conditions using a common reference point. In mature markets, benchmarks often become embedded in contracts, reporting systems, analytics platforms and risk management frameworks.

Bid

A bid is an expression of buying interest. It represents the price level at which a market participant is willing to purchase a commodity under specific conditions.

Because bids frequently contain restrictions related to volume, quality, timing, logistics or payment terms, they should not automatically be interpreted as guaranteed transactions. Nevertheless, bids provide valuable information about demand, liquidity and market sentiment.

Offer

An offer is an expression of selling interest. It represents the level at which a market participant is willing to sell a commodity under specified conditions.

Offers help define the upper side of the market range and contribute to the process of price discovery. Like bids, offers may contain commercial conditions that affect their executability.

Executed Trade

An executed trade is a completed transaction between buyer and seller. Among all market signals, executed trades generally provide the strongest evidence of market value because commercial agreement has already been reached.

However, a single executed trade does not automatically represent the broader market. Special logistics conditions, unusual volumes, distressed sales or unique quality specifications may make a transaction less representative than it initially appears.

Indicative Price

An indicative price is a non-binding market indication.

Indicative levels help market participants understand where others currently see the market, but they do not necessarily represent executable opportunities.

Indicatives are particularly useful in fragmented OTC markets where continuous transaction data may not be available. They provide orientation rather than confirmation.

Respondent

A respondent is a market participant who contributes pricing information to a benchmark process.

Respondents may include traders, exporters, processors, brokers, elevators, producers and local buyers. Each respondent observes only a portion of market activity, but collectively they help create a representative picture of the market.

The quality of a benchmark depends heavily on the diversity, activity and credibility of its respondent network.

Liquidity

Liquidity describes how easily a commodity can be bought or sold without significantly affecting the market price.

High liquidity is usually characterized by active participation, frequent transactions, multiple counterparties and relatively narrow spreads. Low liquidity often results in wider spreads, greater uncertainty and more volatile price behavior.

In physical commodity markets, liquidity often appears in clusters rather than as a continuous stream.

Thin Market

A thin market is a market with limited trading activity and a small number of meaningful observations.

In thin markets, individual bids, offers or transactions can exert disproportionate influence on market perception. This increases the risk that unusual observations may be mistaken for representative market behavior.

Because of this, benchmark methodologies often apply stricter publication criteria in thin markets.

FOB (Free On Board)

FOB is one of the most widely used delivery bases in international commodity trade.

Under FOB terms, the seller is responsible for delivering the commodity onto a vessel at a designated export port. Once loading is completed, responsibility transfers to the buyer.

FOB prices therefore incorporate the costs, risks and logistics required to move the commodity from its origin to the export vessel.

CPT (Carriage Paid To)

CPT is a delivery basis under which the seller pays for transportation to a specified destination.

Examples include CPT port, CPT elevator, CPT processor and CPT terminal. Because CPT reflects a different stage in the logistics chain than FOB, the two prices often differ substantially. Understanding CPT values is essential for evaluating local procurement economics and transportation efficiency.

Freight

Freight refers to the cost of transporting commodities through a supply chain. Freight may include trucking, rail transportation, river logistics, ocean shipping or combinations of multiple transportation modes.

Changes in freight costs frequently influence basis relationships, export competitiveness and regional price formation. During periods of logistical stress, freight can become one of the most important drivers of physical market behavior.

Spot Market

A spot market is a market where commodities are traded for near-term physical delivery.

Unlike futures markets, spot markets focus on execution, logistics, quality, timing and immediate supply-demand conditions.

Most agricultural spot markets operate through decentralized networks of participants rather than centralized exchanges, which makes structured benchmarks particularly valuable.

Futures Market

A futures market is a centralized marketplace where standardized contracts are traded for future delivery or settlement.

Futures markets facilitate price discovery, risk transfer and hedging. They primarily reflect expectations about future conditions rather than immediate physical realities.

Although futures markets strongly influence physical markets, they do not fully describe local execution conditions, logistics constraints or regional supply-demand imbalances.

Price Discovery

Price discovery is the process through which markets determine value.

In physical commodity markets, price discovery occurs through a combination of bids, offers, negotiations, transactions, logistics information, supply-demand dynamics and market expectations.

Effective price discovery requires both information and liquidity. Benchmarks help improve price discovery by transforming fragmented observations into structured references.

Spread

A spread is the difference between two related prices. Examples include FOB-CPT spreads, regional basis spreads, processing spreads and transportation spreads.

Spreads often provide more information than outright prices because they reveal changes in logistics, liquidity, demand and market structure. Experienced commodity professionals frequently monitor spreads as closely as headline prices.

Market Center

The market center is the price area that best represents the balance of buying and selling interest at a given moment.

It does not necessarily correspond to the highest bid, the lowest offer or the most recent transaction.

Most benchmark methodologies are designed to identify the market center rather than individual observations.

Outlier

An outlier is an observation that differs significantly from the rest of a sample.

Outliers may result from unusual market conditions, data errors, special transactions or non-representative commercial situations.

A robust benchmark methodology evaluates outliers carefully rather than automatically accepting or rejecting them.

Median

The median is the middle value in an ordered dataset.

Unlike an arithmetic average, the median is less sensitive to extreme observations and unusual values. This makes it particularly useful in fragmented markets where occasional outliers may appear.

For many commodity benchmarks, the median provides a more stable representation of the market center than a simple average.

Outlier Filter

An outlier filter is a methodological tool used to identify observations that differ substantially from the broader sample.

Its purpose is not to remove inconvenient data but to protect the benchmark from distortion caused by unusual, erroneous or non-representative observations.

Outlier filters improve benchmark stability and credibility.

Price Reporting Agency (PRA)

A Price Reporting Agency is an organization that collects, evaluates and publishes benchmark price assessments.

PRAs typically rely on transparent methodologies, respondent networks and validation procedures to produce market references.

Many globally recognized commodity benchmarks are produced by PRAs.

Market Intelligence

Market intelligence is the process of transforming market information into actionable understanding.

Information alone does not create intelligence. Intelligence emerges when data is organized, compared, interpreted and placed into context.

Modern commodity market intelligence combines benchmark data, logistics information, basis analysis, historical trends, liquidity indicators and market events to support decision-making.

Market Infrastructure

Market infrastructure consists of the systems, methodologies and institutions that allow markets to function efficiently.

Examples include exchanges, benchmarks, settlement systems, reporting platforms, APIs, analytics tools and data services.

As commodity markets become increasingly digital, benchmarks are evolving from simple publications into foundational infrastructure layers that support broader ecosystems of analytics, automation and decision support.

API (Application Programming Interface)

An API is a structured mechanism that allows software systems to exchange information automatically.

In commodity markets, APIs enable benchmarks, analytics platforms, dashboards, trading systems and risk management tools to access market data in machine-readable form.

APIs transform benchmark data from static publications into operational components of market infrastructure.

Market Transparency

Market transparency is the degree to which participants can understand market conditions, pricing behavior and liquidity dynamics.

Transparency does not require disclosure of every transaction. Instead, it requires sufficient information for participants to evaluate market conditions fairly and consistently.

Well-designed benchmarks improve transparency while preserving commercial confidentiality.

Market Infrastructure Layer

A market infrastructure layer is a set of systems that supports market operation without directly participating in trading activity.

Benchmarks, reporting frameworks, APIs, settlement systems and analytics platforms can all function as infrastructure layers.

These layers help transform fragmented markets into more structured, comparable and efficient environments.

Reference Point

A reference point is a commonly accepted market value used for comparison, negotiation and analysis.

Benchmarks function as reference points because they provide a shared frame of reference for participants who may otherwise observe very different portions of the market.

Without reference points, market communication becomes fragmented and difficult to scale.

About the Projects

This handbook was developed within the broader ecosystem of projects focused on commodity market infrastructure, analytics and benchmark development.

1D3X

1D3X develops local commodity index infrastructure designed to help markets create transparent, methodology-driven benchmarks.

The goal is to build a global network of local commodity indices capable of supporting analytics, contracts, APIs and future market infrastructure.

https://1d3x.com

MN7R

MN7R focuses on market intelligence, analytics and decision-support systems for commodity markets.

Its objective is to transform fragmented market signals into structured information that can improve decision quality.

https://mn7r.com

Cr0pto

Cr0pto explores the future of index-based and programmable commodity markets.

The project investigates how benchmark infrastructure can support tokenized and digitally native commodity trading environments.

https://cr0pto.com

Together, these initiatives share a common objective: *to make commodity markets more transparent, more structured and more understandable.*